

Article

Simulation-Based Optimization of Crane Lifting Position and Capacity Using a Construction Digital Twin for Prefabricated Bridge Deck Assembly

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Abstract: The growing adoption of off-site construction methods has increased the critical role of mobile cranes within the construction sector. This study develops a Construction Digital Twin (CDT) framework to optimize crane lifting positions and capacities for the installation of prefabricated bridge decks. By integrating 3D site modeling, Building Information Modeling (BIM), and crane simulations within the Unity game engine, the CDT overcomes the limitations of conventional 2D-based planning by providing a three-dimensional representation of site conditions. An exhaustive search method identifies optimal crane configurations, enhancing precision and efficiency. Simulation calibration using video analysis of real bridge deck installations aligns crane speed and cycle times with actual operations, improving reliability. Case studies demonstrate the CDT's ability to reduce crane operation costs by 27% when employing a smaller capacity crane while maintaining operational efficiency. Additional DFA-focused simulations with varying deck dimensions revealed a potential 10% cost reduction by optimizing crane operations and deck design strategies. The CDT framework supports early-stage planning, reduces operational risks, and contributes to cost-effective and safer construction practices, offering a scalable solution adaptable to various construction scenarios.



Academic Editor: Humberto Varum

Received: 17 December 2024

Revised: 28 January 2025

Accepted: 28 January 2025

Published: 3 February 2025

Citation: Jang, D.-H.; Roh, G.-T.; Jeon, C.-H.; Shim, C.-S. Simulation-Based Optimization of Crane Lifting Position and Capacity Using a Construction Digital Twin for Prefabricated Bridge Deck Assembly. *Buildings* **2025**, *15*, 475. <https://doi.org/10.3390/buildings15030475>

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Keywords: off-site construction (OSC); construction digital twin (CDT); prefabricated deck; crane operation; BIM; exhaustive search; construction optimization; design for assembly (DFA)

1. Introduction

1.1. Background

The construction industry has faced persistent challenges, including low productivity, limited innovation, and inadequate predictability [1]. To address these issues, the adoption of advanced digital technologies has become a critical strategy. Among these technologies, Digital Twin (DT) is recognized for its capability to provide precise digital simulations that can improve various stages of construction projects [2]. A Digital Twin is defined as a digital replica of a physical object or system that enables real-time monitoring, simulation, and optimization throughout the construction lifecycle. Its implementation supports decision-making processes, predictive maintenance, and operational efficiency [3]. The integration of Building Information Modeling (BIM) with real-time data from Internet of Things (IoT) devices has further advanced the application of DTs in construction. This combination

enables the creation of comprehensive digital representations of physical assets, facilitating improved data interoperability and real-time monitoring [4].

In crane operations, DTs have demonstrated practical applications by enabling the simulation and optimization of crane systems. This approach has been shown to improve safety, efficiency, and productivity in construction site operations. Despite its advantages, the adoption of DTs in construction is constrained by challenges such as data integration, interoperability between systems, and the absence of standardized frameworks. Addressing these challenges is essential for the wider adoption and effective use of DT technology in the industry [5]. In summary, the integration of Digital Twins with BIM and IoT technologies provides an approach to enhance productivity, safety, and efficiency in construction processes, including crane operations. Ongoing research continues to address existing barriers, paving the way for the broader implementation of DTs in the construction industry [6].

The prefabricated deck method, a key component of off-site construction (OSC) in bridge projects, is increasingly favored for its efficiency and reduced on-site duration. However, implementing this method, particularly in complex or large-scale projects, introduces substantial challenges. Traditional planning for prefabricated deck installations using mobile cranes depends on 2D drawings and crane charts. This approach necessitates skilled engineers to evaluate variables such as crane lifting positions, working radius, and crane capacity. While adequate for straightforward sites, these methods often fall short in complex three-dimensional settings like bridge construction, which may require the use of multiple or high-capacity cranes. In such instances, engineers typically resort to conservative planning with extensive safety margins to avoid risks, potentially leading to the underutilization of crane capabilities and resulting inefficiencies. Conversely, underestimating risks can escalate the likelihood of accidents or project delays. Therefore, balancing risk management and operational efficiency is essential in complex construction settings.

Previous research on mobile crane operations in construction has largely concentrated on site interference checks utilizing Building Information Modeling (BIM) [7] and crane movement paths. However, these strategies do not fully address the complexities of on-site crane operations during prefabricated deck installations, where the interplay between site constraints, crane capacities, and safety issues demands a dynamic analysis.

This study introduces a Construction Digital Twin (CDT) framework that incorporates 3D site modeling, bridge BIM models, and crane motion programming to enhance construction planning. The CDT facilitates a simulation-based evaluation and optimization of the precast deck slab division plan and the corresponding construction methods using mobile cranes. By creating an interactive virtual environment, the CDT allows for more accurate planning that adapts to complex site conditions and optimizes crane operations. The simulation provides detailed insights into optimal crane lifting positions and load capacities, thereby enhancing safety and efficiency in crane operations and reducing the risks of accidents and delays associated with conventional methods.

1.2. Problem Statement

In planning the erection of prefabricated bridge decks using mobile cranes, traditional methods primarily depend on 2D drawings and generalized crane charts. However, these tools, combined with the experiential knowledge of planning engineers, often fail to accurately reflect detailed, site-specific conditions that are crucial for precise crane operations. This shortfall results in erection plans that may be overly conservative, leading to the underutilization of equipment and operational inefficiencies, or, conversely, the underestimation of risks, potentially causing safety hazards and project delays. Currently, there is a lack of advanced, intelligent tools that effectively address the complexities of crane operation planning for such projects.

This study addresses these challenges by developing a predictive CDT that integrates 3D site modeling, bridge BIM models, and mobile crane modeling and operations into a unified interactive platform. By meticulously defining each component and establishing their interconnections within the Unity engine, the CDT facilitates a detailed simulation of crane operations that considers site-specific conditions and project requirements. This comprehensive approach aims to enhance planning accuracy, optimize crane utilization, improve safety measures, and ultimately streamline construction processes. The development of the CDT marks a significant step in addressing the limitations of conventional methods and acts as a catalyst for further digitalization in site construction.

As outlined in the KOSHA guide for mobile cranes [8], the working load of a truck crane equipped with outriggers must not exceed 85% of its critical load. The lifting load at the working radius must be determined using the lifting load chart, while the boom configuration is selected based on vertical height considerations in the boom combination diagram. Moreover, the crane's lifting capacity must be verified under the condition of fully extended outriggers. However, real-world site constraints, such as limited space for outriggers and the presence of obstacles, often restrict the available crane lifting positions. Additionally, existing structures, including girders and previously installed prefabricated decks, can impede boom movement or object handling. Collision detection is thus a critical consideration that cannot be addressed adequately using 2D drawings alone. These site-specific limitations may also reduce the safety margin required for crane operations.

The inadequacies of traditional planning methods often lead to overly conservative plans, resulting in the underutilization of equipment and operational inefficiencies. Conversely, an underestimation of risks can pose serious safety hazards and cause project delays. Despite these challenges, there is a notable absence of advanced tools capable of effectively addressing the complexities inherent in crane operation planning for prefabricated bridge deck erection. To overcome these limitations, this study highlights the importance of incorporating detailed 3D site models and bridge BIM models equipped with collision detection functionalities for simulating prefabricated deck erection.

This research proposes the development of a predictive Construction Digital Twin (CDT) that integrates 3D site modeling, bridge BIM models, and mobile crane modeling into a unified, interactive simulation platform. By systematically defining each component and its interdependence within the Unity engine, the CDT enables detailed simulations of crane operations that account for site-specific conditions and project constraints.

In the existing crane lifting plan, there is a lack of consideration for optimizing the crane lifting position. The traditional method involves setting the crane's position and checking whether the crane's working radius, based on its capacity, covers the center of the object to ensure safety. In this study, the exhaustive search method is applied to evaluate all potential positions, enabling the optimization of the crane's position. This approach aims to enhance planning accuracy, optimize crane utilization, improve safety protocols, and streamline construction processes. Furthermore, the development of this CDT represents a pivotal advancement in addressing the limitations of conventional methods, paving the way for further digitalization and innovation in site construction practices.

1.3. Research Objectives

The primary goal of this research is to develop a predictive Digital Twin (DT) for the erection of prefabricated bridge decks, optimizing crane operations by integrating essential components. This DT utilizes precise 3D site modeling derived from Unmanned Aerial Vehicle (UAV) data, modified BIM for crane management, and detailed crane operation modeling. The objective is to improve crane performance, thereby enhancing decision-making, boosting safety, and minimizing operational risks.

The research focuses on the following specific objectives: “3D Site Modeling”, “BIM Modeling and Manipulation”, and the “Realization of Mobile Crane Modeling and Operation”.

As for 3D Site Modeling, utilizing UAV data, 3D site models were constructed to integrate BIM and crane operation simulations. These models accurately depict the terrain and site constraints, employing UAV photogrammetry, and are refined through techniques such as mesh reconstruction and point cloud analysis to ensure high precision. As for BIM Modeling and Manipulation, the BIM has been adapted to isolate individual components, such as prefabricated bridge decks, which can be stored in a trailer and lifted by a mobile crane. This modification facilitates the simulation of crane operations within the DT framework and optimizes the handling of prefabricated decks. As for the Realization of Mobile Crane Modeling and Operation, mobile crane operations have been meticulously modeled to simulate essential functionalities, including the interplay between boom length, working radius, and lifting capacities as dictated by crane charts. The model incorporates critical safety features such as collision detection to prevent unsafe scenarios and ensure a realistic portrayal of crane activities. This comprehensive modeling supports precise crane maneuvering, lifting, and placement under various site constraints. It further aids in optimizing lifting positions and capacities through systematic approaches like the exhaustive search algorithm. By identifying the optimal crane configurations for diverse site conditions, the model significantly enhances the efficiency and safety of the prefabricated deck erection process. Integration of DT Components: Integrating 3D site models, modified BIM components, and mobile crane models into a unified CDT involves considerable complexity. A platform that can seamlessly combine these elements and facilitate real-time interactions is crucial for effective simulations. The Unity engine offers a viable solution, but implementing it entails overcoming challenges related to data compatibility, synchronization, and control of interactive simulations.

Calibration with Real-World Data: Precise calibration is crucial to ensure that the CDT accurately mirrors real-world conditions. This process involves overcoming computational limitations and validating simulation results against actual or expected outcomes. The calibration utilized real crane operation data derived from video footage, which helped fine-tune the simulation to accurately reflect crane speed, lifting time, and operational safety. **Case Studies for Optimization:** The CDT underwent testing through various case studies that included simulations with different crane capacities, deck dimensions, and site conditions. These studies demonstrated the CDT’s capability to offer data-driven recommendations for optimizing crane lifting positions and enhancing construction efficiency. By achieving these objectives, the CDT acts as a comprehensive tool for simulating and improving crane operations during the erection of prefabricated bridge decks, thereby contributing to safer and more efficient planning.

2. Literature Review

2.1. Digital Twin Applications in Construction

The construction industry has increasingly adopted DT technology to enhance project outcomes by synchronizing physical and digital systems. Originally developed in the aerospace and automotive industries to improve product lifecycle management by providing comprehensive digital representations of physical assets [9,10], DT has been recognized for its potential to improve safety, efficiency, and decision-making throughout the project lifecycle [11]. However, the application of DT technology in construction often involves the integration of advanced technologies such as artificial intelligence (AI), machine learning (ML), and the Internet of Things (IoT), which demand significant computational resources

and expertise, thus limiting their scalability and accessibility for small- to medium-scale projects [12,13].

DT technology involves creating a dynamic digital representation of a physical asset or system, enabling real-time monitoring, simulation, and optimization [9]. In construction, DTs enhance decision-making by providing current information on project status, resource utilization, and potential risks [11]. However, despite DTs' potential, many studies focus on integrating advanced technologies, which may not be feasible for all construction projects due to their high cost and complexity [14].

2.2. Site Modeling for CDT

Accurate site modeling is essential for the successful implementation of a CDT. UAV-captured point-cloud data are being increasingly used to create detailed 3D models of construction sites, owing to their high accuracy and efficiency [15,16]. Integrating this point-cloud data into game engines like Unity facilitates immersive and interactive simulations, thereby improving the visualization and analysis capabilities of DTs [17]. However, converting point clouds to mesh models and optimizing them requires advanced technical skills and can be time-consuming, which may limit accessibility for smaller projects.

The precision of a 3D site model generated from UAV-captured images is influenced by various factors, including terrain characteristics and image processing methods. In a case study focusing on a beach area by Lee et al. (2016) [18], UAV-based modeling achieved a horizontal accuracy of ± 58 mm in the X direction and ± 47 mm in the Y direction, while the vertical error was ± 86 mm. This level of accuracy meets the standards for allowable errors in cadastral and topographic surveys, even in challenging conditions such as sandy terrains with insufficient image matching points. Therefore, for construction projects like bridges or roads, 3D site modeling using UAV-captured photos is sufficiently accurate for simulating project activities.

2.3. BIM Modification and Utilization in CDT

BIM acts as a cornerstone for integrating various data streams within CDT applications. BIM provides a robust framework to encapsulate both geometric and semantic details of construction projects, enhancing collaboration among stakeholders and improving decision-making processes [19]. However, traditional BIM models often lack the necessary flexibility for direct simulation, especially in the crane-based handling of prefabricated components [20]. To address this, modifications to BIM components are essential for accurately simulating real-world construction activities, such as the storage, transport, and lifting of prefabricated elements, thus making them compatible with CDT integration.

2.4. Crane Modeling for DT Applications

Optimizing crane operations is crucial for enhancing efficiency and safety in construction projects, particularly in large-scale operations like bridge construction. Previous research has investigated various optimization algorithms and simulation methods for planning crane operations; however, these methods typically require specialized expertise and significant computational resources, which can limit their scalability [20,21]. This study adopts a practical and cost-effective approach—the exhaustive search method—to optimize crane operations without the need for advanced computational inputs.

2.5. Integration of DFMA in CDT

Furthermore, design for manufacturing and assembly (DFMA) principles have been incorporated into the CDT framework to refine the design and assembly of prefabricated components. By simulating different deck dimensions and configurations, the influence of design decisions on crane lifting capacities, working radii, and assembly processes

was assessed. This analysis facilitates the creation of optimal designs that harmonize manufacturing efficiency with on-site assembly restrictions. Integrating DFMA principles with crane operational planning ensures that prefabricated components are not only easy to manufacture but also straightforward to assemble with the available crane resources [22].

2.6. Identified Gaps and Comprehensive Integration of CDT for Construction

Despite significant advancements in DT technology within the construction industry, numerous critical gaps persist. Many studies have concentrated on integrating sophisticated technologies such as AI, ML, and IoT. However, the application of these technologies may not be feasible for all construction projects due to their high costs and complexity [14]. Moreover, the scalability of DT applications is often hampered by the requirement for advanced computational resources, restricting their use in smaller- and medium-scale projects [19].

Previous DT research has largely focused on enhancing specific areas of construction management, such as real-time monitoring and predictive maintenance, through the use of these advanced technologies. However, these approaches frequently target specific tasks, limiting their widespread application and scalability across the broader industry. This study seeks to address these limitations by developing a CDT that optimizes crane operations for erecting prefabricated bridge decks without an overreliance on high-tech solutions.

While 3D site models are widely used for visualization, their application as operational platforms for simulating crane repositioning and optimizing lifting positions is limited. This limitation restricts the ability to dynamically address site-specific constraints during construction planning. Conventional BIM frameworks primarily focus on static representations and lack the flexibility needed for the dynamic manipulation of individual components. Existing studies do not adequately explore the use of BIM models where prefabricated components, such as bridge decks, can be detached, stored, and handled by cranes. This gap hinders the integration of BIM into comprehensive planning tools for construction operations. Despite the growing adoption of prefabrication methods, research specifically addressing the erection of prefabricated bridge decks is scarce. Most studies focus on general prefabrication strategies and do not fully address the unique challenges of crane operations, such as lifting positions, site-specific spatial constraints, and equipment requirements. The optimization methods employed in previous research often rely on heuristic or computationally intensive approaches. These methods may not align with the practical conditions of construction sites.

Employing an exhaustive search method, this research offers a practical and cost-effective approach that enhances accessibility and scalability for a broad range of construction projects. The exhaustive search method systematically evaluates all possible crane configurations to determine the optimal setup that satisfies lifting requirements while minimizing operational risks. This method takes into account various factors including crane capacity charts, working radii, and site constraints to ensure that the selected crane configurations are both feasible and efficient without necessitating extensive computational resources.

Unity, a widely accessible game engine, was utilized to integrate crane models and BIM data for simulation purposes. By developing custom scripts and utilizing Unity's robust physics engine, the crane operations, lifting processes, and interactions with prefabricated components were effectively simulated. This integration furnishes an interactive platform that allows for the real-time visualization and adjustment of crane positions, significantly enhancing decision-making during the planning phase without the need for specialized software or hardware.

In summary, this research has delivered a comprehensive CDT application customized for the erection of prefabricated bridge decks, optimizing crane lifting positions, crane capacity, and overall construction logistics. This study aims to boost construction performance in terms of cost efficiency by simulating and planning all relevant activities. The proposed solution addresses a significant gap by providing an accessible, construction-focused CDT application that supports all stages of prefabricated bridge deck erection, contrasting with previous studies that often focused on isolated tasks or specific technologies.

3. Methodology

3.1. Overview of Research Design

This section details the methodology employed to develop a CDT aimed at optimizing crane operations during the prefabricated bridge deck erection process. The principal goal of this research is to identify the optimal crane lifting positions (P^*) and crane capacities (C^*) that maximize the total number of successfully installable bridge decks (n). This objective can be mathematically expressed as Equations (1) and (2)

$$(P^*, C^*) = \underset{P, C}{\operatorname{argmax}} n, \quad (1)$$

$$n = f(WR_C, BL_{\text{Sitecondition}}, P_{\text{targetDeck}}, P_{\text{crane}}) \quad (2)$$

where

WR_C : maximum working radius for specific crane lifting capacity C
 $BL_{\text{site condition}}$: required maximum boom length for each site condition
 $P_{\text{target deck}}$: farthest position of installable deck
 P_{crane} : crane lifting position

The constraints in this optimization problem fall into three primary categories: accessibility, deck installation position, and equipment. Each category is elaborated below. Accessibility constraints arise from site conditions, which might include barriers such as roads, streams, or slopes that restrict access to the crane and trailer. To ensure efficient operations, the crane and trailer must be positioned only in locations deemed accessible, considering the presence of these obstacles. This ensures that the crane's base remains on stable and safe ground throughout its operations. Formally, this constraint is represented as Equation (3)

$$(x_g, y_g, z_g) \notin \text{Obstacles} \quad (3)$$

where (x_g, y_g, z_g) denotes the position of the crane, and "Obstacles" refers to any physical barriers or hazardous areas within the construction site.

Deck installation position constraints relate to ensuring that the crane's position aligns with the deck installation sequence. The crane must be capable of placing each deck at its designated target location according to a predetermined erection order. Additionally, spatial constraints stem from the bridge structure itself, as the crane must avoid positioning directly beneath the bridge deck during operations to maintain safety. These constraints are defined by the original and target positions of each deck, specified by the coordinates and dimensions of the respective decks in Equations (4) and (5).

$$\text{Sequential Installation: } D_i \text{ for } i = 1, 2, \dots, n \text{ installed in order} \quad (4)$$

$$\text{Crane Working Boundary: } d(P_c, D_i) \leq R_c, i = 1, 2, \dots, n \quad (5)$$

where D_i indicates i th deck installed in order. P_c and R_c are the fixed crane position and the crane working radius, respectively. $d(P_c, D_i)$ represents the distance between the crane position P_c and deck D_i , ensuring that each deck is within the working range of the crane.

Equipment constraints relate to the crane's technical specifications, including limits on boom length, working radius, and lifting capacity. To successfully install each prefabricated deck, the crane's boom length must be sufficient to reach the required height, and its lifting capacity must be capable of handling the deck's weight within a specified working radius. The crane lifting position is determined based on the relationship between the horizontal center of the crane and the pivot center of the farthest deck. This distance must fall within the crane's working radius to ensure successful operation. The relationship can be expressed mathematically as Equation (6).

$$\|P_C - P_D\| \leq R_{\max} \quad (6)$$

where P_C is the crane lifting position (horizontal center of the crane), P_D denotes the pivot center of the farthest deck, and $R_{\max}$ denotes the maximum working radius of the crane. The term $\|P_C - P_D\|$ represents the Euclidean distance between the crane's horizontal center and the pivot center of the farthest deck. Given the target location of this deck, the crane's lifting position can be adjusted by measuring the maximum working radius, which is determined based on the boom length and the weight of the object being lifted.

Furthermore, during crane operation, neither the crane itself nor any object attached to it can come into contact with the bridge structure or any other obstacles. These constraints are effectively concerning in regard to collision. Mathematically, a collision can be represented by an intersection test between the geometries of two objects, often based on their boundaries or bounding volumes. If we define two objects, A and B , in 3D space with volumes V_A and V_B , a collision occurs, and it can be expressed as Equation (7).

$$V_A \cap V_B \neq \emptyset \quad (7)$$

where $\cap$ denotes the intersection of the volumes. This intersection implies that the two objects occupy the same space within a defined environment, indicating a collision. Collision detection is a computational process used to identify intersections or contacts between objects in a simulation. Conceptually, this involves continuously monitoring the positions and boundaries of objects to ensure that no unintended overlaps occur, which is essential for accurate and safe CDT simulations. These constraints ensure that the crane operates effectively while adhering to safety standards.

Due to the complexity of these constraints, finding an analytical solution to the optimization problem of crane positioning and capacity is impractical. The construction environment is highly dynamic and involves numerous variables and unpredictable factors that complicate optimization. Therefore, the use of CDT is essential. The CDT provides a virtual representation of the construction environment, allowing for an exhaustive exploration of possible crane positions and capacities. The methodology developed in this research integrates several key components to create a comprehensive virtual environment: 3D site modeling, BIM, and mobile crane modeling. A 3D site model, generated from real-world site data, accurately represents the terrain and any obstacles, ensuring a faithful reproduction of the actual conditions. The BIM captures the prefabricated bridge deck components, which are then manipulated and integrated into the Digital Twin environment.

In this study, we employed the exhaustive search method to determine the optimal crane lifting positions and capacities. This method systematically explores all possible configurations to identify the most effective solution. Although this approach is computationally demanding, it guarantees precision in crane operations and ensures the selection

of the most efficient crane configuration for the erection of prefabricated bridge decks. The CDT enables detailed simulations of crane movements and positioning, providing critical insights that are vital for construction planning and enhancing overall efficiency.

To ensure the Digital Twin accurately reflects real-world conditions, we incorporated a calibration process into the research framework. This calibration involves using video recordings from actual crane operations to validate the accuracy of the CDT, ensuring that the simulations closely match on-site operations. By integrating real-world data into the virtual model, the CDT becomes an invaluable tool for planning and optimizing crane operations, offering a cost-effective solution that significantly enhances both safety and efficiency in the erection process of prefabricated bridge decks.

This study develops a comprehensive methodology that integrates 3D site modeling, BIM, and mobile crane operational modeling within a virtual environment. Utilizing an exhaustive search method, the study identifies optimal crane lifting positions and capacities, ensuring precision in crane operations for the erection of prefabricated bridge decks. This method facilitates a detailed simulation of crane movements and positions, providing valuable insights into construction planning and enhancing efficiency. This scalable and practical approach offers a cost-effective solution for optimizing crane operations during construction projects. The research framework incorporates real-world calibration, using video recordings of actual crane operations to ensure that the CDT accurately reflects real-world activities.

The research has developed a CDT framework specifically aimed at optimizing crane lifting positions and capacities during the prefabricated bridge deck erection process. This methodology integrates UAV-based 3D site modeling, BIM manipulation, and crane operation simulations within the Unity game engine, providing a practical, scalable, and cost-effective solution for crane operation optimization without relying on complex AI/ML technologies. The key components of the research design include 3D site modeling, bridge BIM manipulation, mobile crane modeling, and operational programming. CDT combined three different components: site 3D model, bridge BIM model, and equipment model. One of the key factors that enables the CDT to operate for the prefabricated deck installation simulation is the integration. In this research, Unity engine was used as a platform on which the three components can be unified operation. In order to generate the CDT, modeling data format should be FBX or OBJ file format, which can be imported into Unity platform. After being imported into the platform, each model can be manipulated to form the CDT. Figure 1 provides a schematic overview of the research design.

3.2. 3D Site Modeling Using UAV Data

The initial step in developing the CDT involves creating an accurate 3D site model. UAVs are utilized to capture high-resolution aerial images of the construction sites. These images are processed using photogrammetry to generate a 3D point cloud, which is subsequently converted into a detailed site model using Bentley iTwin 2022 (Bentley Systems, Exton, PA, USA) [23] platform. Figure 2 illustrates the image-processing operation for creating the 3D site model from UAV-captured photographs. Several green dots depicted above the site in Figure 1 represent the UAV positions from which the photographs were taken.

This 3D model accurately represents the terrain and physical constraints of the construction site, thereby providing a foundation for subsequent integration with other components. The generated 3D model was essential for precise planning because it allowed for the identification of site-specific conditions, such as the location of obstacles, site access points, and terrain slopes, all of which influence crane operations. Generating a 3D site model is a multistep process involving various techniques and software tools. The procedure for creating a 3D site model, which is a key component of DT construction, involves sequential

activities. The process begins with capturing high-resolution aerial photographs of the construction site using UAVs. The UAVs flew over the site and captured overlapping photographs to ensure comprehensive coverage. Once the photos were captured, they were entered into the iTwin platform.

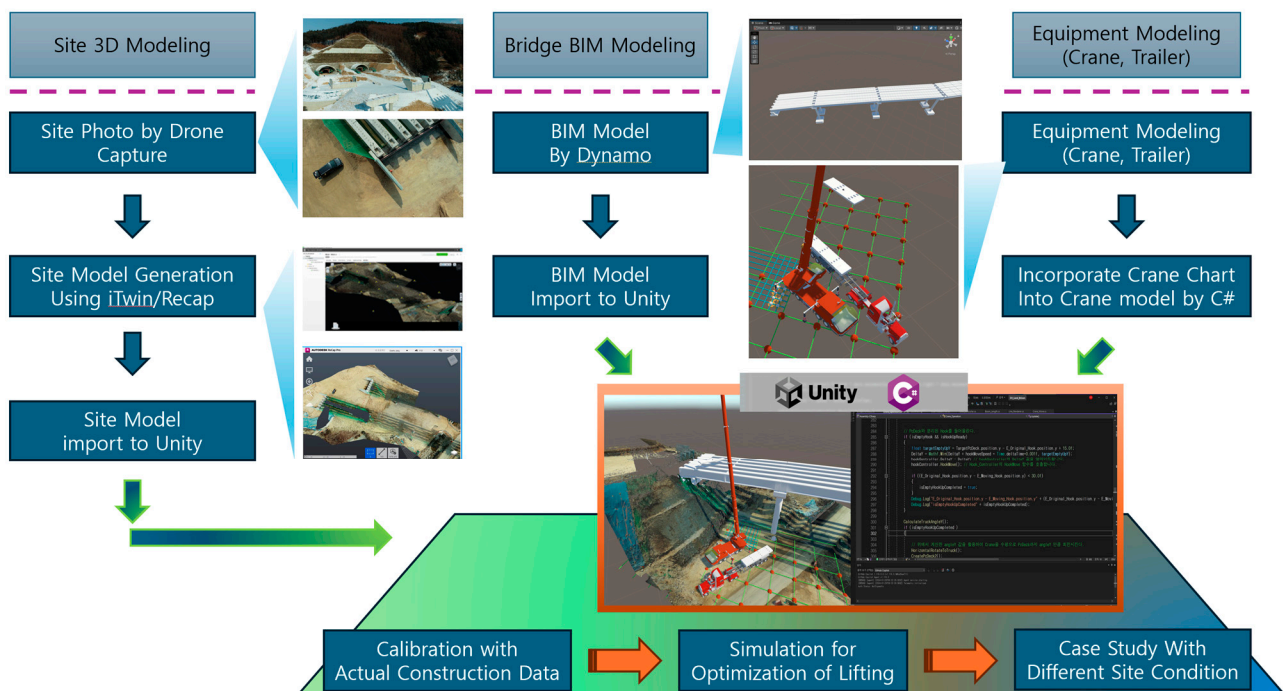


Figure 1. Overview of the Research Design.

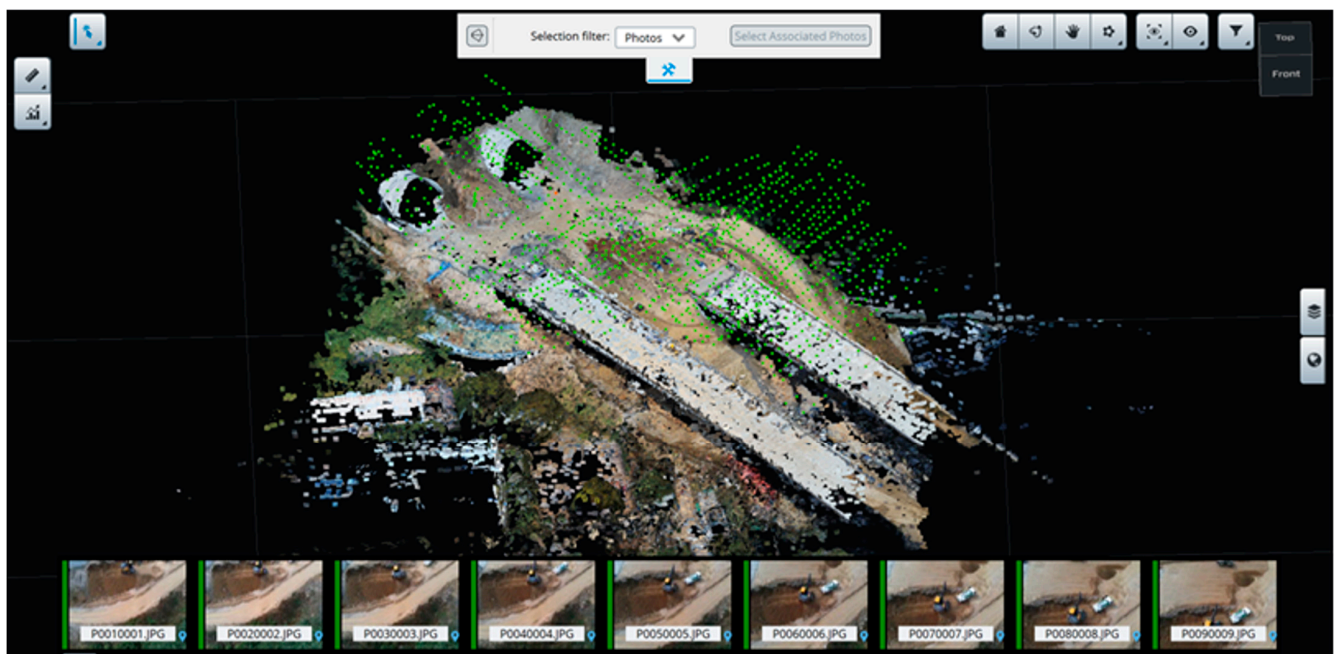


Figure 2. Generation of 3D site model using UAV-captured images.

iTwin processed the images and prepared them for further analysis. Aerotriangulation was performed on the input photographs to establish precise spatial relationships between the images. This step involves calculating the coordinates of the points on the ground and creating a network of control points. Reconstruction was performed to convert 2D images

into 3D models. This process involves the generation of a point cloud, which is a collection of data points representing the 3D surface of a site. The reconstruction process produces a point-cloud model. This model consists of millions of data points that accurately represent the geometries of construction sites. The point cloud model was edited using Autodesk Recap 2022 (Autodesk Inc., San Rafael, CA, USA) [24] software. This step involves removing unnecessary points and improving the accuracy of 3D representation. The edited point cloud is used to reproduce the 3D model in a mesh format. A mesh is a network of vertices, edges, and faces that define the shape of a 3D object. The 3D mesh model was imported into the Unity 2022.3.23f (Unity Technologies, USA) [25] in the FBX format. This format ensures that the model retains detailed geometry and texture information.

In Unity, mesh components are combined to form a cohesive 3D model representing the complete construction site. A mesh collider is assigned to this 3D model to enable accurate physical interactions within the virtual environment, facilitating collision detection and other simulations. Additionally, grid and crane-lifting points are integrated into the model to assist in the precise positioning and planning of crane operations on the construction site. The 3D site modeling undergoes a rigorous verification process to ensure its accuracy and completeness. This process confirms that the model accurately reflects a real-world site and meets all project requirements. Following verification, the 3D site model is integrated with bridge BIM and crane modeling, culminating in a systematic approach for generating 3D site models, as depicted in Figure 3.

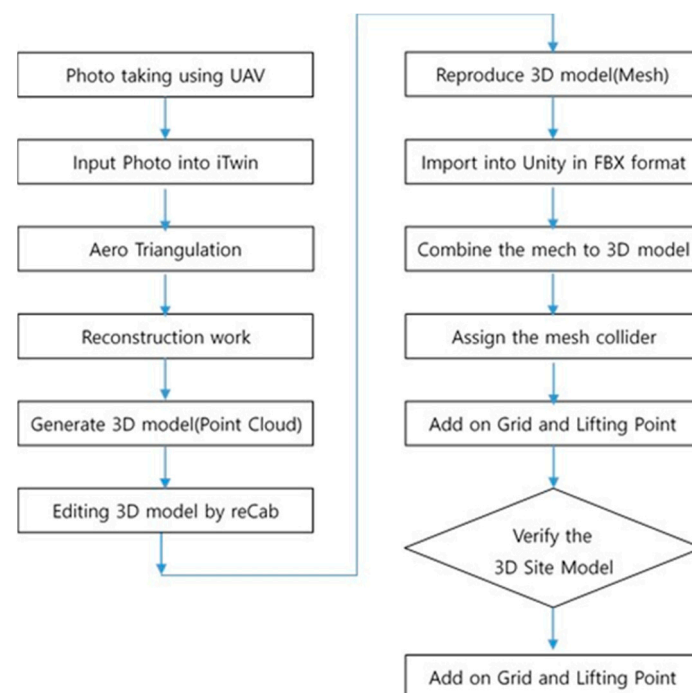


Figure 3. Procedure for generating a 3D site model.

To balance computational efficiency with spatial accuracy, 5 m grid system was selected after evaluating several alternatives. This interval allows for realistic site modeling while minimizing processing demands. Although a narrower grid could provide greater precision, it would also increase processing time and computational resource requirements. Starting with 5 m interval enables the identification of an initially rough optimal location, which can then be refined for higher accuracy as needed. Figure 4 illustrates the grid system generated on the 3D site map, featuring grid lines at 5 m intervals and numbering to identify specific positions. The baseline of the grid begins from the edge of the bridge deck, acknowledging that crane placement directly beneath the bridge is infeasible. Similarly, perpendicular

grid lines start from the edge of the slope to avoid unstable terrain unsuitable for crane positioning. The grid is drawn using a relative coordinate system that distinguishes accessible locations from obstacles or inaccessible areas. Due to the high resolution of 3D site maps, obstacles such as slopes, roads, and streams are clearly classified, facilitating accurate planning for crane placement.

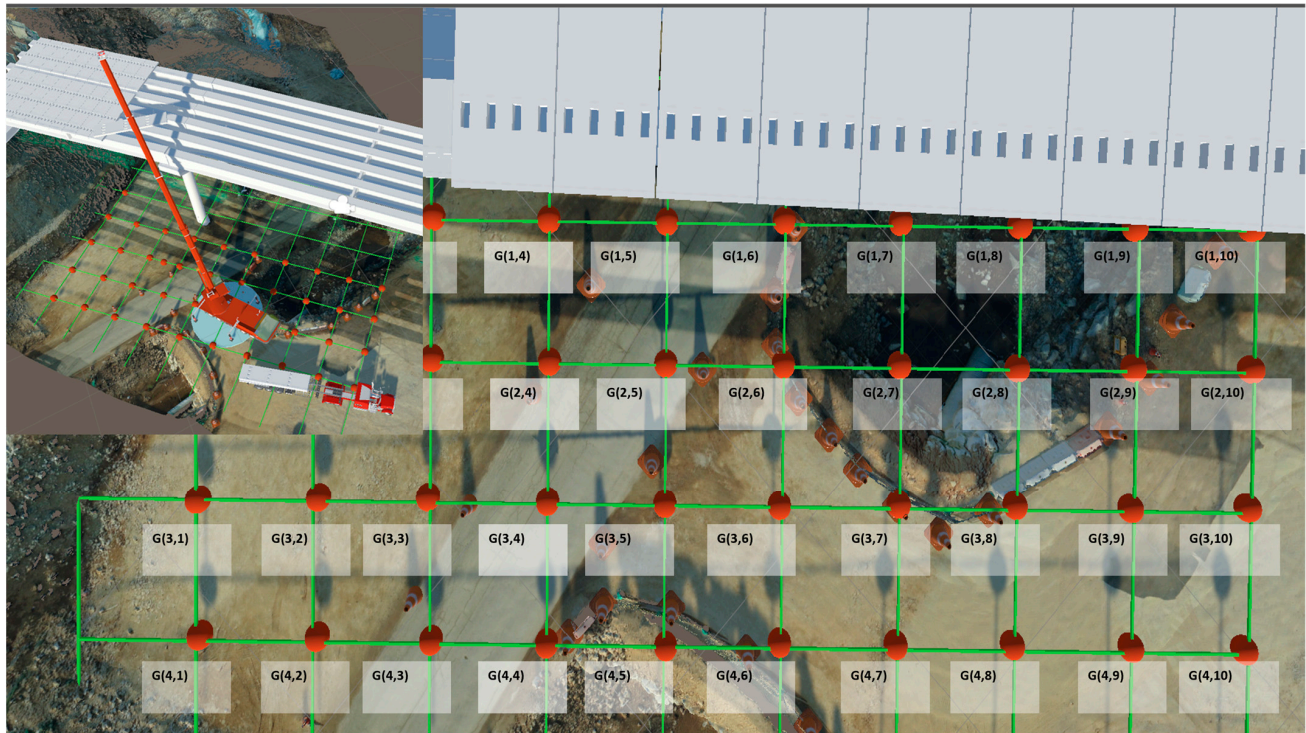


Figure 4. Grid generation on the 3D site map.

Table 1 illustrates the site conditions and status, mapped according to grid numbers. This table highlights crane positions that have been filtered based on site-specific constraints such as slopes, obstacles under the bridge, roads, and limitations due to the working radius.

Table 1. Filtering based on site conditions.

Items (Inaccessible Position for Crane)	Grid No	Remarks
Slope	G(1,1), G(2,1), G(4,1)	
Under Bridge	G(1,2), G(1,3), G(1,4)	
Road	G(1,5), G(2,4), G(2,5), G(3,3), G(3,4), G(3,5), G(4,3), G(4,4)	to be allowed to access by decision-making
Obstruction (Stream)	G(1,6), G(1,7), G(1,8), G(1,9), G(1,10), G(2,6), G(2,7), G(2,8), G(2,9), G(3,7), G(3,8), G(4,5), G(4,6)	temporary platform or filling required for accessibility
Out of Working Radius	G(4,7), G(4,8), G(2,10), G(3,9), G(3,10), G(4,9), G(4,10)	subject to crane capacity

3.3. BIM Modeling and Manipulation

BIM plays an essential role in the CDT by representing bridge structures and pre-fabricated deck components. The BIM was adapted to support crane simulations, with each prefabricated deck modeled as a distinct entity that can be lifted and assembled by the crane. This modification is critical for enabling realistic simulations of crane operations. The bridge BIM model was integrated with a 3D site model using the Unity engine to create an interactive simulation environment. The modification of the BIM facilitates component-level interactions, allowing prefabricated decks to be detached from the main

bridge model and manipulated within the CDT. This adaptability is crucial for simulating various crane lifting positions and evaluating different assembly strategies. The deck installation sequence for bridge BIM involves a systematic approach to ensure precise and efficient crane operation. This section outlines the procedures for establishing the deck installation sequence on the Unity platform as follows:

- **Renaming Deck Objects According to Installation Sequence and Work Volume:** Each deck object in the hierarchy is renamed according to the installation sequence and work volume, as defined by the crane operation with a fixed lifting position. This renaming ensures clarity and organization during the installation process.
- **Programming Crane Operation Sequence:** The crane operation sequence is programmed using the Crane.cs script, where the prefix character denotes the work volume associated with a specific lifting position, and the number following this prefix indicates the installation sequence. In this research, deck naming adheres to the model inventory system proposed by Nguyen et al. (2022) [26]. Therefore, each deck is labeled as “SP#_DK#”, with the sequences for the first span named “SP 1_DK 1”, “SP 1_DK 2”, ..., “SP 1_DK 31”, “SP 1_DK 32”. In this setup, the designated deck is moved from its storage location to the trailer according to the installation sequence.

Prefabricated deck erection activities vary significantly when using a mobile crane, due to the erection sequence. The decks are erected according to a predetermined sequence. Thus, if the first deck is beyond the working radius, the subsequent erection activities cannot proceed. Figure 5 illustrates the concept of deck erection, showing Decks 1, 2, and 4 as being outside the working radius, which prevents their erection in this work zone. If the erection sequence is adjusted to exclude Decks 1, 2, and 4, the remaining decks can be erected according to the planned sequence.

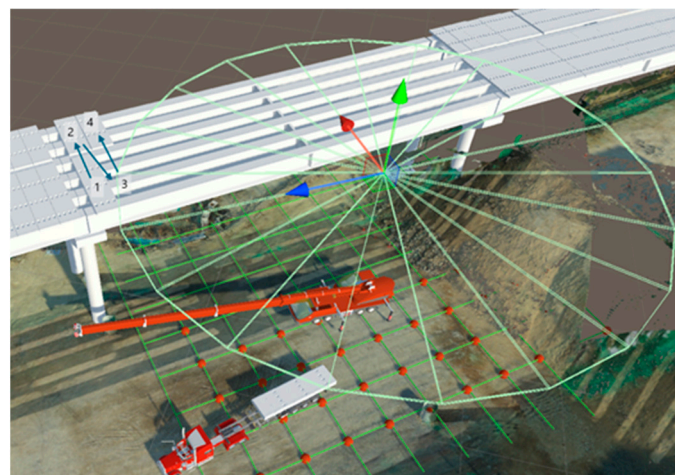


Figure 5. Concept of deck installation.

3.4. Mobile Crane Modeling

Mobile crane modeling is a crucial aspect of the CDT, as crane operations significantly influence the efficiency and safety of bridge deck erection. A crane model was developed to simulate key functions such as boom extension, rotation, and lifting capacities, aligning with actual crane specifications. The model incorporates crane charts to determine allowable lifting capacities based on boom length and working radius. Collision detection was also implemented to ensure safe crane operations within the simulated environment (Figure 6).

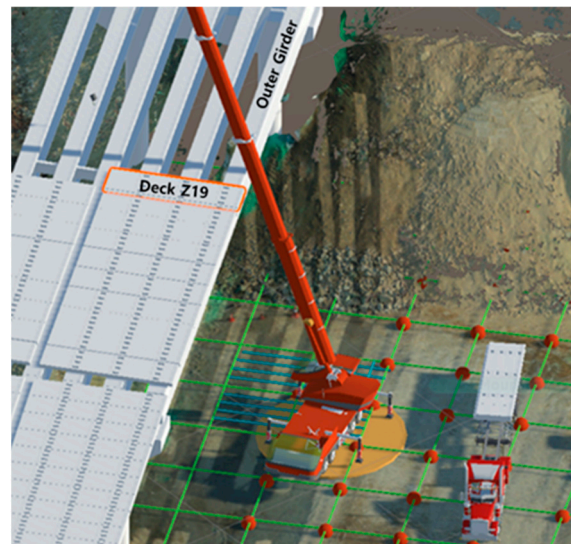


Figure 6. Collision detection against the outer girder during deck erection.

The crane model is programmed to detect potential collisions with the bridge structure, site obstacles, or other prefabricated components, thereby minimizing risks during the planning phase. This proactive approach allows for the optimization of crane lifting positions, enhancing both safety and operational efficiency.

3.4.1. Incorporation of Crane Chart into Mobile Crane Modeling

Crane charts are indispensable tools for planning heavy-lifting operations with mobile cranes. Manufacturers provide these charts for specific crane models, detailing their lifting capacities and operational parameters. Integrating a crane chart into a CDT is a strategic approach to optimizing crane operations. The crane chart encompasses three primary factors: working radius, boom length, and weight of the load. The weight of the object to be lifted, such as a prefabricated deck, is determined by its dimensions. The boom length is calculated based on site conditions, specifically the difference in elevation between the crane's lifting center and the height of the target object. From the crane chart, one can readily determine the working radius for a given crane based on its boom length and the weight of the object.

In conventional practices for planning crane lifting activities, a circle is drawn at the center of the crane lifting position. If the object to be lifted falls within this circle, the crane's lifting capacity is deemed sufficient for the task. However, the longer the boom length, the smaller the working radius for lifting the same weight. Thus, if the object lies outside this circle due to the boom length, the crane's position must be adjusted closer to the object, or a crane with a larger capacity should be selected. In DT modeling, the boom length and working radius can be adjusted to modify the crane's lifting capacity based on the crane chart.

The crane model within the DT was meticulously programmed to incorporate the crane charts provided by manufacturers. These charts define critical parameters such as working radius, boom length, and lifting capacity, which are vital for planning crane operations. The integration of crane charts into the CDT allows for real-time assessment of operational limits based on the weight of the deck, the required lifting height, and boom extension. Simulation of crane behavior under various configurations to optimize lifting performance, while visualization of crane capacity utilization across different positions ensures both safety and efficiency. The CDT utilizes this detailed information to generate

operational constraints, dynamically adjust crane configurations, and assist site engineers in selecting the most appropriate crane and setup for each specific operation.

Despite the usability and practicality of crane charts, they have inherent limitations due to their static nature and lack of real-time adaptability. Since crane charts provide predefined lifting capacities based on ideal conditions, they do not dynamically account for site-specific constraints such as terrain irregularities, obstacles, or limited space for outrigger deployment. This static nature can lead to overly conservative planning, where engineers allocate excessive safety margins to avoid potential risks, often resulting in underutilization of crane capacity and increased operational costs. Conversely, in complex environments, relying solely on crane charts without real-time adaptability may lead to underestimated risks, as critical spatial constraints or unexpected site conditions are not reflected in the static lifting capacities. To overcome these limitations, the CDT framework integrates crane charts within a 3D simulation environment, allowing for dynamic assessments rather than relying solely on static reference data. Unlike traditional crane planning, which assumes generalized lifting conditions, the CDT incorporates detailed 3D site models that account for terrain features, restricted zones, and spatial constraints, ensuring a realistic evaluation of crane operations. By simulating lifting positions in a virtual construction site, the CDT provides an interactive planning tool that adapts crane configurations based on real-world constraints.

Furthermore, the CDT framework enhances lifting feasibility by incorporating collision detection algorithms that evaluate potential interferences between crane components, bridge structures, and previously installed decks. Unlike traditional crane charts, which do not account for obstructions, the CDT applies mesh colliders and box colliders to detect possible collisions in real time. This capability enables site engineers to identify potential risks early and make necessary adjustments, ensuring that lifting operations are both safe and efficient. By integrating crane charts with 3D site modeling and real-time collision detection, the CDT overcomes the static and non-adaptive nature of conventional crane charts. These enhancements enable a more precise, flexible, and efficient method for determining crane lifting positions and capacities, ultimately improving safety and construction workflow efficiency.

To facilitate the installation sequence and crane operation, several C# scripts are developed and attached to the corresponding objects within the Unity environment. These steps ensure a well-organized and efficient setup for the deck installation sequence, enabling precise control over crane operations and facilitating smooth installation of prefabricated components. By following this procedure, the construction process is optimized, leading to improved accuracy and reduced risk of errors. General operation of crane function can be implemented in C# script *Crane.cs*. As for the manipulation of the BIM component, with which BIM component can be handled in line with mobile crane, apart from the BIM manipulation algorithm *Meshpositionfixer.cs*, *Loader.cs* and *Operator.cs* algorithm were additionally programmed. *Loader.cs* enables decks to be stacked prefabricated decks on the trailer. Trailer bed can be acknowledged as the original deck position. *Operator.cs* has the function of assigning the installation sequence and the working boundary. Pseudo algorithms for crane operation are shown in Algorithm 1.

3.4.2. Enhanced Collision Detection

To ensure operational safety, the CDT incorporates advanced collision-detection algorithms. These algorithms use mesh colliders for irregular surfaces, supplemented by box colliders around the deck components to establish safety margins. This approach enables the system to identify potential clashes between crane components and bridge structures.

It also assesses the feasibility of crane movement in constrained spaces, provides visual alerts, and recommends alternative configurations in the event of a detected collision.

Algorithm 1. Crane Operation: Crane.cs

1. Initialize Crane Components:
 - Find and assign crane parts: horizontal axis, vertical axis, boom, hook.
 - Detach the hook initially.
 2. Main Update Loop:
 - Check if any movement trigger flags are active.
 - Horizontal Axis Rotation: Start rotating the horizontal axis to a target
 - Vertical Axis Rotation: Start rotating the vertical axis to a target
 - Boom Length Adjustment: Start adjusting the boom length
 - Hook Height Adjustment: Start adjusting the hook to reach the target.
 3. Simulate Hook Movement:
 - Calculate hook position based on boom end position and wire length.
 - If an object is grabbed, update its position to match the hook's.
 4. Component Rotation and Adjustment:
 - Horizontal Axis Rotation: Gradually rotate, checking for collision.
 - Calculate expected rotation to align with target.
 - Vertical Axis Rotation:
 - Calculate expected rotation angle for target height.
 - Boom Length Adjustment:
 - Calculate required boom length to reach target. /Adjust boom length
 - Hook Height Adjustment:
 - Calculate wire length to reach target. /Adjust wire length
 5. Collision Detection:
 - Use a sphere cast to check for any obstacles
 - If a collision is detected, stop the current movement
 6. Grab and Release Object:
 - Grab an object by attaching it to the hook.
 - Release the grabbed object by detaching it from the hook.
 7. Visualize Crane Movements:
 - Visualize crane components and collision detection
-

Collision detection was implemented using a ray casting method, which projects a spherical ray with a defined radius in the 3D environment to effectively detect obstacles during crane operations. This method was formulated into a procedural algorithm to support the broader goal of optimizing crane lifting positions in Digital Twin (DT) simulations. The use of sphere casting instead of standard ray casting provides a crucial safety margin by accounting for the physical dimensions of the crane boom and its surroundings. This ensures more thorough collision detection, particularly in scenarios where precision and safety are critical for safe crane operations. The algorithm is designed to efficiently identify potential collisions by projecting a virtual sphere along the crane boom's movement path, ensuring safe operations in a simulated construction environment. Algorithm 2 shows the pseudo code for collision detection using the ray casting method.

Algorithm 2. Ray casting method for collusion detection in crane operation

1. Initialize Boom Direction and Distance:
 Compute the direction vector between the crane boom's starting point (Boom.position) and its endpoint (BoomEnd.position).
 Determine the distance between these points to calculate the length of the boom.
2. Perform Collision Detection Using SphereCast:
 Use a sphere casting operation from the crane boom as the origin, with a specified collision radius (BoomCollisionCheckRadius) to project a spherical ray along the calculated direction.
 If an obstacle is detected within the defined range, trigger an exception (CraneCollisionDetectedException) to prevent unsafe movement.

3.4.3. Synchronizing BIM Components with Crane Operations

A significant accomplishment of this research was the synchronization of BIM components with crane operations. The BIM model, containing detailed geometries of the prefabricated decks and structural elements, was enhanced to allow dynamic interactions with the crane model. Key modifications included:

- Assigning pivot centers to each deck for accurate lifting simulation.
- Incorporating collision-detection mechanisms to prevent clashes during lifting operations.

The pivot center is defined as the central point around which an object in a digital model rotates, scales, or transforms. Unlike traditional BIM, where static representations of structures are prevalent, the pivot center introduces a dynamic operational capability. This concept allows for accurate control over transformations and movements, which is essential in complex simulations such as crane operation or prefabricated component assembly within a Digital Twin environment. The pivot center is essential to enable precise and reliable simulation of construction activities, such as crane operations, within the digital environment. It is the key concept to achieving dynamic manipulation of BIM components, ensuring each component can be positioned, rotated, and installed with the necessary level of control. This capability is critical to the study's objective of optimizing crane lifting positions and enhancing the practicality of Digital Twin simulations. In this research, the pivot center is implemented through programming components that adjust each object's pivot according to its intended interactions. Using Unity, for example, scripts such as MeshPositionFixer.cs assign pivot coordinates to structural components, ensuring alignment and stability during simulated construction activities. This implementation supports accurate handling, positioning, and assembly of BIM elements, allowing more realistic and practical simulations of on-site operations. The assignment of pivot centers on each deck should be considered an essential step in BIM manipulation for data delivery. Establishing pivot points during the BIM model creation phase will facilitate seamless data transfer into simulation environments, such as Digital Twins. This research proposes that BIM models incorporate pivot-setting tools, enabling engineers to predefine rotation and movement centers within the model. Such advancements would improve interoperability between BIM and simulation platforms, reducing the time needed for additional adjustments and enhancing the accuracy of dynamic operations in Digital Twins.

The integration of BIM components, particularly the prefabricated deck elements, with crane operations in Unity involves a series of steps. This section outlines the procedure for synchronizing the movement of BIM components with crane operations to ensure smooth construction simulations. The first step involves importing the bridge BIM model into Unity. This step ensures that all prefabricated deck components are accurately represented within the simulation environment. Next, the inspector settings for the BIM model are adjusted. This includes enabling read/write mode to allow modifications and interactions with the model in Unity.

A mesh collider [25] is attached to each prefabricated deck component. This attachment is crucial for enabling physical interactions and collision detection between the deck components and other elements in the Unity environment. A mesh position fixer, implemented as a C# script, is attached to each prefabricated deck component. This script ensures that the position of each component is accurately fixed and can be manipulated as required during crane operations. The scene within Unity is then switched on to generate an empty object. This empty object serves as a placeholder and reference point for positioning the prefabricated deck components. The modified bridge BIM modeling, with all necessary adjustments and attachments, is converted into a Prefab [25]. A Prefab in Unity is a reusable asset that can be replicated and manipulated within the scene. After creating the Prefab, the scene [25] is switched off. This step is necessary to finalize the modifications and prepare for the next steps in the process. The original bridge BIM modeling is then deleted from the scene. This deletion ensures that only the modified Prefab, with all required functionalities, remains for further use. Finally, the modified bridge BIM modeling Prefab is reallocated within the hierarchy of the Unity scene. This reallocation positions the Prefab appropriately, ensuring that it moves in coordination with crane operations. By following these steps, the BIM components, particularly the prefabricated deck elements, are enabled to move in synchronization with crane operations within the Unity environment. This integration enhances the accuracy and efficiency of simulating and managing construction processes in the CDT framework. These adjustments enable individual decks to detach from the main BIM structure and align with crane lifting parameters. Figure 7 illustrates the sequence of manipulating a BIM component in conjunction with crane movement.

This synchronization ensured seamless integration of BIM with crane operations within the CDT, facilitating real-time simulation of lifting paths, rotation angles, and deck placement. The alignment of crane motion with BIM components creates a realistic and interactive environment for assessing operational feasibility.

The pivot center is defined as the central point around which an object in a digital model rotates, scales, or transforms. Unlike traditional BIM, where static representations of structures are prevalent, the pivot center introduces a dynamic operational capability. This concept allows for accurate control over transformations and movements, which is essential in complex simulations such as crane operation or prefabricated component assembly within a Digital Twin environment. In this research, the pivot center is implemented through programming components that adjust each object's pivot according to its intended interactions. Using Unity, for example, scripts such as `MeshPositionFixer.cs` assign pivot coordinates to structural components, ensuring alignment and stability during simulated construction activities. This implementation supports accurate handling, positioning, and assembly of BIM elements, allowing more realistic and practical simulations of on-site operations. The pivot center is introduced to enable precise and reliable simulation of construction activities, such as crane operations, within the digital environment. It is central to achieving dynamic manipulation of BIM components, ensuring each component can be positioned, rotated, and installed with the necessary level of control. This capability is critical to the study's objective of optimizing crane lifting positions and enhancing the practicality of Digital Twin simulations. Algorithm 3 shows the pseudo code for `FixMeshPosition.cs`, which enables us to match geometry center to pivot center for each deck component.

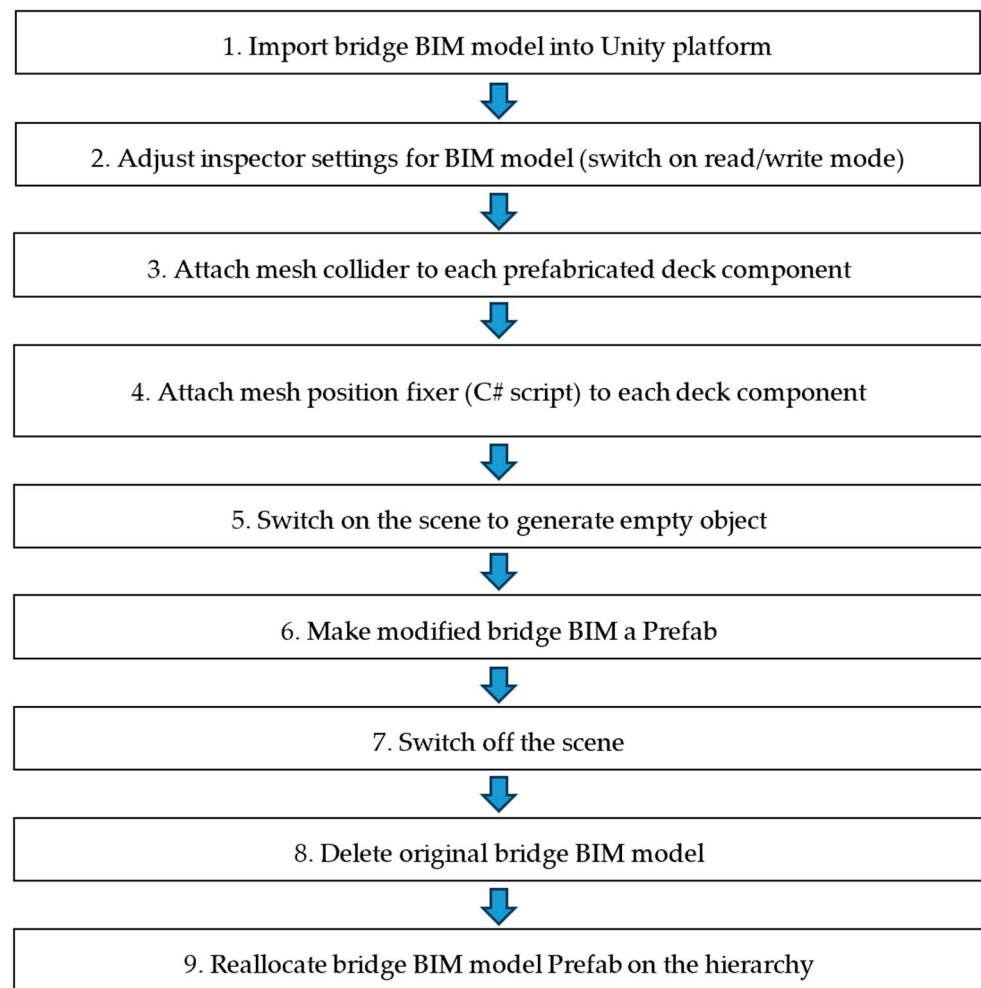


Figure 7. Manipulation of a BIM component in coordination with crane movement.

Algorithm 3. Fixing Mesh Position

Input: MeshFilter, MeshCollider

Output: Mesh with Center at (0, 0, 0)

Procedure FixMeshPosition()

1. Get the MeshFilter component of the object.
2. Calculate the center of the mesh bounds (center).
3. Adjust the local position by adding center and multiplying by the local rotation.
4. Set the local rotation of the transform to identity (Quaternion.identity).
5. Calculate the offset to shift the mesh center to (0, 0, 0).
6. For each vertex in the mesh, do:
 - Apply the offset to shift the vertex.
 - Rotate the vertex using the original local rotation.
7. Create a new mesh (newmesh) and assign the updated vertices.
8. Copy the triangles, UVs, and colors; recalculate bounds, normals, and tangents for newmesh.
9. Save the new mesh as an asset in the project directory.
10. Replace the original mesh with the new mesh in MeshFilter.
11. If a MeshCollider component exists, replace its shared mesh with the new mesh.
12. Destroy the MeshPositionFixer script component.

End Procedure

Data synchronization between different components was achieved by establishing common data exchange standards and ensuring compatibility across platforms. This

integration enabled real-time interaction, allowing the crane to be operated within the 3D site model to evaluate various lifting strategies and their impacts on construction efficiency.

3.5. Integration of DT Components

The integration of the 3D site model, BIM components, and mobile crane model was a pivotal step in the development of the Construction Digital Twin (CDT). Unity was selected as the primary platform for this integration due to its robust real-time rendering capabilities, advanced physics simulation, and support for interactive user interfaces. The integration process involved multiple stages, each addressing a specific component of the CDT, as detailed below.

The 3D site model was created based on real site data captured using UAV (Unmanned Aerial Vehicle) technology. UAV-captured photos were processed using Bentley iTwin 2022 (Bentley Systems, USA) [23] for image processing, followed by point cloud editing in Recap [24]. The processed point cloud data were then converted into OBJ/FBX formats using iTwin [23] to produce a detailed and scalable 3D site model. The bridge's BIM model was developed using Dynamo API within Autodesk Revit 2022 (Autodesk Inc., USA) [27] as the BIM authoring tool. This enabled the creation of a detailed representation of the prefabricated components and their subsequent adjustment for crane handling and operation. The mobile crane model was developed using operational parameters derived from crane charts and method statements. The modeling process utilized the Unity game engine [25] to create a functional crane model integrated with the CDT. The specific tools and adjustments applied during this process are summarized in Table 2. The final integration combined the 3D site model, the adjusted bridge BIM model, and the mobile crane object within Unity. The components were seamlessly connected to form a cohesive and interactive CDT, enabling real-time simulation and visualization of construction activities. This integration supports construction planning and optimization by providing a detailed and dynamic virtual environment.

Table 2. Integration of the Digital Twin with 3D site, bridge BIM, and crane modeling.

Description	Input Data	Tools and Method	Output
Site Modeling	Real Site	UAV	UAV Captured Photos
	UAV Captured Photos	iTwin: Image Processing Recap: Point Cloud Editing iTwin: Point Cloud Convert to OBJ/FBX Format	3D Site Modeling
Bridge BIM Modeling	Drawings (2D)	Dynamo: BIM Authoring Tool	BIM Modeling
	Bridge BIM	C# Programming: MeshPositionFixer.cs Unity Game Engine C# Programming : Crane.cs : Loader.cs : Operator.cs	Adjusted 3D BIM Object for Crane Handling
Mobile Crane Modeling	Crane Chart Method Statement		Automated Operating Game Object
Integration	Site Modeling Adjusted Bridge BIM Mobile Crane Object	Unity Game Engine C# Programming	Construction Digital Twin

Through this integration, the CDT offers a unified platform for simulating various construction scenarios, enhancing decision-making, and improving planning efficiency.

3.6. Calibration with Real-World Data

To enhance the accuracy of the CDT, the simulation was calibrated using real-world data derived from video recordings of the prefabricated deck erection activities. Parameters such as crane hook speed, rotation angles, and lifting times were extracted from these

recordings and applied to the virtual crane model. This calibration process ensured that the CDT accurately reflected actual crane operations, rendering the simulation output both realistic and reliable for planning purposes. Accurate data on crane operation were obtained by video recording the prefabricated deck installation process. These recordings provided essential data for analyzing crane operations and installation procedures, thus supporting the simulation of these activities within a real-world context.

To assess crane operations, the distance moved by the crane hook during deck installation was meticulously measured. The movements of the deck were categorized into vertical and horizontal curved movements, resulting from the crane's swing actions. Figure 8 presents a schematic of the deck movements and distances covered during the installation process.

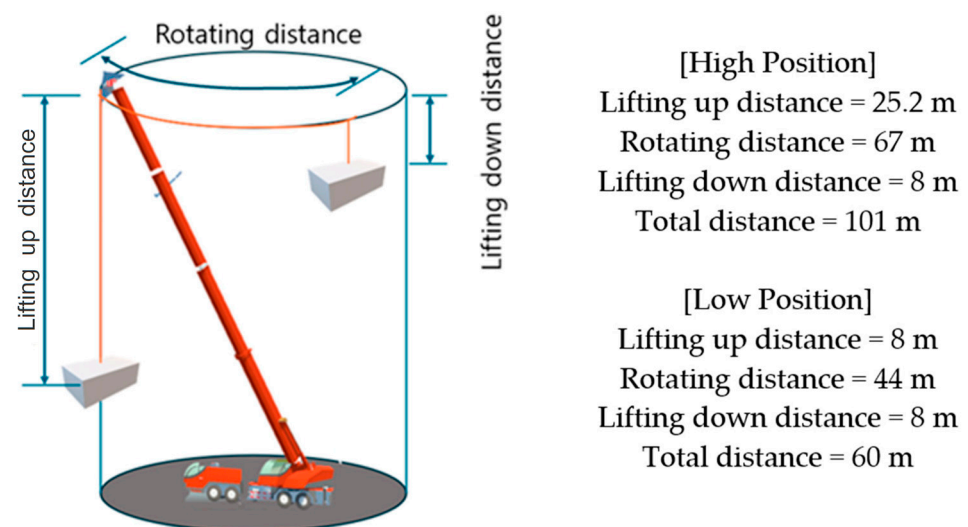


Figure 8. Schematic representation of deck movement and distance covered.

A detailed analysis of the video data was conducted to capture precise timings for each phase of the crane operations. This analysis revealed variations in the execution of identical activities under consistent site conditions, suggesting that the conditions and installation coordinates did not significantly differ. It was determined that the key factors influencing crane operation were closely related to the arrangement and preparation of shear anchors and their associated activities.

Deck lowering and installation are pivotal activities in the construction of prefabricated bridge decks. The crane's capacity and lifting position significantly influence overall performance, often more so than the actual operation time of the crane. To enhance crane performance, key variables for crane operations were determined based on existing site conditions. These calculated parameters lay a solid foundation for the planning and execution of prefabricated deck erection. On the Unity platform, the crane operation speed was fine-tuned by adjusting the time scale in the project settings to ensure that the simulation aligned with real-world crane operation data. The calibrated CDT was then utilized to conduct several test scenarios, comparing crane configurations, lifting positions, and deck dimensions under various site conditions. Figure 9 illustrates the pattern of deck erection activities. The graph indicates that horizontal movement without an increase in distance represents idle time caused by the setup or repair time of shear anchors. By analyzing the slope for each activity, crane operation parameters such as lifting speed, rotation speed, and lowering speed can be determined.

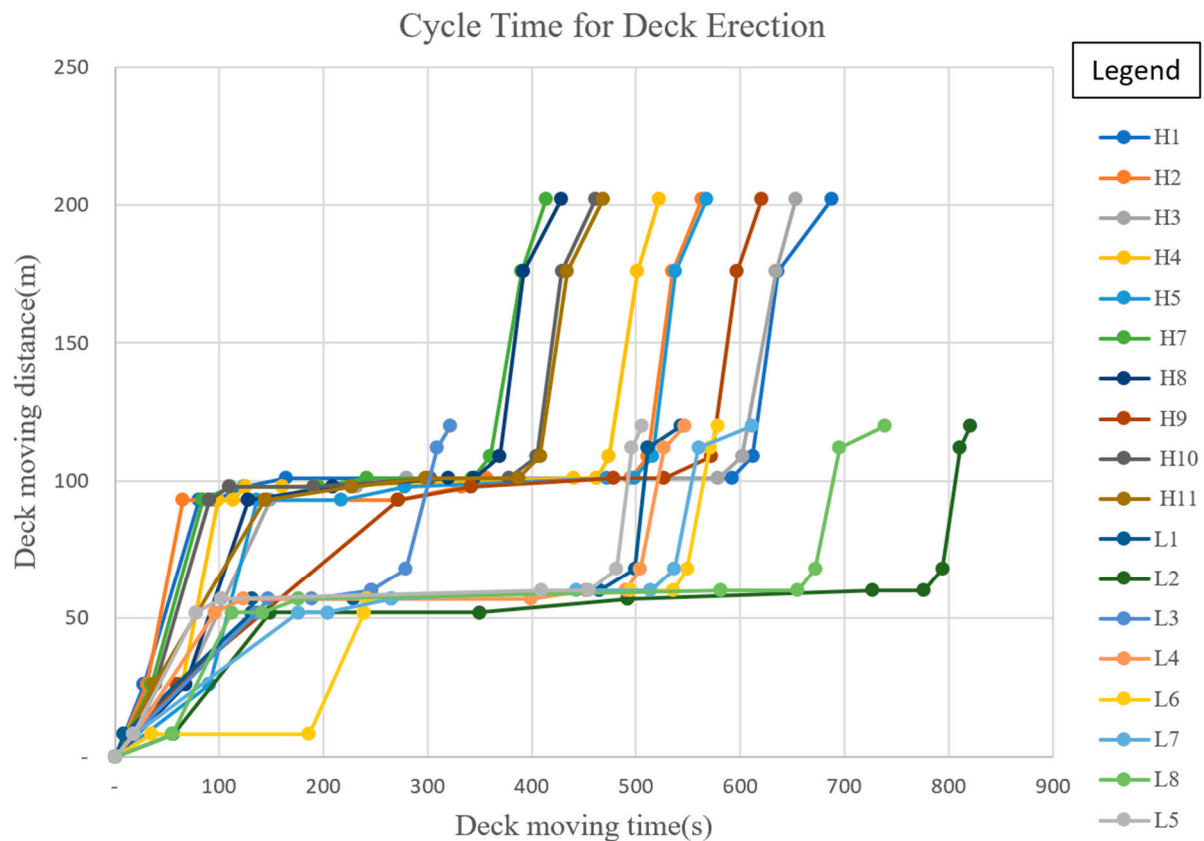


Figure 9. Cycle time for deck erection.

Table 3 presents the crane operation cycle time for high boom height conditions. In each case, the critical path is determined by the process of lowering the deck onto the girder and its subsequent installation.

Table 3. Crane operation schedule (high boom height).

[High Boom Height]				(Unit: s)
Activity	Min	Max	Average	Remark
PC Deck Lifting Up	28	91	53	
Crane Rotating with PC Deck	37	181	72	
PC Deck 1st Lowering	26	70	49	
PC Deck 2nd Lowering and Installation	40	136	102	Critical Path
PC Deck Unhooking and Lifting Up	7	20	22	
Crane Rotating without PC Deck	30	24	26	
Crane Lifting Down	24	52	30	

In prefabricated bridge deck erection, deck lowering and installation are critical activities. The crane's capacity and lifting position play a more significant role in the overall performance than the actual operation time of the crane. To accurately integrate crane operations into CDT simulation, key operational variables were derived based on the given

site conditions. These variables were determined using the average crane operation time for each activity. Table 4 presents the calculated input data for mobile crane operation in CDT simulation.

Table 4. Calculation of input data for mobile crane operation.

Item	Calculation Details
Lift Up Speed	Ground Level: 77.3 Deck Top Level: 96.0 Trailer Height: 1.5 m Hook Up Level: + 8.0 m from Deck Top $\therefore$ Lift Up Height = $96.0 - 77.3 - 1.5 + 8 = 25.2$ m : Average Lift Up Time: 53 s. Hook Lift Up Speed = $25.2 \div 53 = 0.5$ m/s
Crane Rotating Speed	Rotating Angle: 120° , Average Rotating Time: 72 s. Rotating Speed: $120 \div 72 = 1.7^\circ/\text{s}$

These calculated parameters provide a solid foundation for planning and executing the prefabricated deck erection. In the Unity platform, the crane operation speed can be adjusted by modifying the time scale in the project settings, ensuring the simulation aligns with real-world crane operation data.

3.7. Constraints for Deck Installation

Several critical constraints must be considered to ensure both the feasibility and safety of crane operations and to optimize the positioning of a mobile crane for installing prefabricated bridge deck objects. These constraints are fundamental to the exhaustive search algorithm used in this study to address the physical and operational limitations inherent in construction environments. The first constraint involves the height of the crane's outriggers relative to the deck installation position is crucial. The z-coordinate of the crane's outriggers must be less than the z-coordinate of all deck installation positions to ensure stability and operational safety. The second constraint ensures that the crane is positioned outside the horizontal bounds of the bridge to prevent potential collisions or interference with the bridge structure, as depicted in Figure 10. The x- and y-coordinates of the crane must not fall within the defined horizontal bounds of the bridge. This constraint is critical for maintaining operational safety and preventing structural damage during the construction process. The third constraint involves obstacle avoidance. The crane must not be positioned over any obstacles or water, as this could compromise the stability and safety of crane operations. This constraint ensures that the crane's base remains on solid and safe ground throughout the operation. Figure 10 illustrates the constraints related to obstacles. Due to the presence of a stream, which is considered an obstacle, the crane can only be positioned adjacent to the stream, despite this location being far from the optimal lifting position.

The final constraint concerns the crane's working radius. The maximum distance from the crane to any deck installation position must not exceed the crane's maximum working radius. This ensures that the crane can reach all necessary points without overstretching or risking structural integrity. These constraints are vital to the optimization process, guiding the selection of feasible crane positions and ensuring that crane operations are both efficient and safe. By adhering to these constraints, the exhaustive search algorithm can effectively determine the optimal crane positioning, facilitating the successful installation of prefabricated bridge deck objects under the given site conditions.

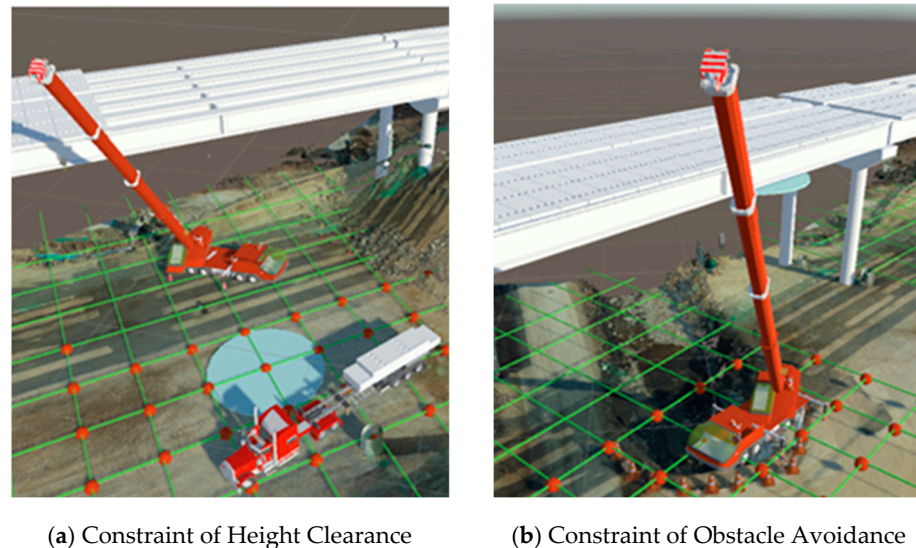


Figure 10. Constraints for crane lifting position.

3.8. Optimization Algorithm

In this study, the optimization of crane-lifting positions and capacities for handling prefabricated bridge deck objects is approached through the exhaustive search method. The primary objective is to determine the optimal crane positions that maximize the number of decks installed from a fixed position while adhering to constraints such as site-specific conditions and varying installation locations. This method systematically evaluates all feasible crane positions, filtering out infeasible options based on predefined constraints. The choice of the exhaustive search method and its comparison with alternative optimization techniques, such as genetic algorithms and linear programming, are discussed to justify its application.

The exhaustive search method was chosen for its simplicity and reliability in addressing constrained optimization problems with a manageable number of grid points [28]. Unlike heuristic methods, such as genetic algorithms, which rely on probabilistic mechanisms and extensive parameter tuning, the exhaustive search method ensures that all potential crane positions are evaluated systematically. This characteristic makes it particularly advantageous for small to medium-sized construction projects, where the computational load remains manageable. The deterministic nature of this method guarantees an optimal solution by exploring every feasible position, a feature that is critical for scenarios where accuracy is essential. For the task at hand, the exhaustive search method effectively handles the challenges posed by crane optimization. It provides a comprehensive feasibility analysis by evaluating all possible crane configurations, ensuring that no viable solutions are overlooked. Feasible grid points are identified through a rigorous filtering process, which accounts for constraints such as crane working radius, site obstructions, and the sequence of deck installations. This method aligns seamlessly with the Construction Digital Twin (CDT) framework, enabling precise simulations and evaluations, and ensures that the optimization results are reliable and actionable.

In comparison, genetic algorithms are well-suited for large-scale optimization problems with complex solution spaces but rely on probabilistic mechanisms that do not guarantee an optimal solution. These algorithms also require extensive tuning of parameters such as mutation rate and population size, which can be challenging to calibrate effectively. On the other hand, linear programming is highly efficient for linear optimization problems with well-defined constraints but struggles to accurately model the nonlinear relationships inherent in crane optimization tasks, such as the interplay between boom

length, lifting radius, and deck weight. In contrast, the exhaustive search method directly incorporates nonlinear constraints without requiring approximations, providing more accurate and realistic results within the CDT framework. Scalability, a potential limitation of the exhaustive search method for larger projects, can be addressed through various strategies. For example, adjusting the grid interval can significantly reduce computational demands while maintaining precision. In this research, a 5-m grid interval was chosen as the initial setup, balancing computational efficiency and spatial accuracy. However, for more precise position detection near the optimal grid boundary, a narrower grid interval of 1 m was applied. This approach effectively refines the search space, avoiding excessive computational resource consumption while pinpointing the optimal crane-lifting position.

Additionally, pre-filtering techniques can be employed to exclude infeasible grid points early in the process, significantly reducing the number of evaluations required. Hybrid approaches can combine exhaustive search with heuristic methods to manage larger search spaces more effectively. For instance, genetic algorithms can identify promising regions of the search space, which can then be refined using exhaustive search for local optimization. Parallel computing offers another solution, allowing multiple grid points to be evaluated simultaneously, thereby reducing computational time. Despite these considerations, the exhaustive search method remains a practical and precise approach for optimizing crane-lifting positions in prefabricated bridge deck erection. Its systematic and deterministic nature ensures comprehensive evaluations and reliable outcomes. By incorporating strategies to enhance scalability, the method can also be adapted to handle larger and more complex construction projects effectively. Future research could further explore hybrid models or advanced computational techniques to extend its applicability.

In conclusion, the exhaustive search method provides a robust solution to the optimization challenges faced in this study. Its integration with the CDT framework ensures precise and actionable results, demonstrating its suitability for the specific scale and requirements of the research. While alternative methods such as genetic algorithms and linear programming have their merits, the exhaustive search method's ability to deliver reliable and accurate solutions makes it the preferred choice for the optimization tasks undertaken here.

3.9. Summary

The methodology outlined in this chapter systematically develops the CDT, emphasizing the integration of 3D site modeling, BIM manipulation, crane modeling, and real-world data calibration. By creating a unified and interactive simulation platform, the CDT emerges as a robust tool for optimizing crane operations in prefabricated bridge construction, thereby enhancing both efficiency and safety.

4. Case Study

4.1. Simulation Setup for Crane Capacity

In the case studies, the CDT proved instrumental in demonstrating the practical application of these advancements. For instance, the deck installation sequence was tested under various site conditions to evaluate its impact on crane repositioning and overall efficiency. The synchronization between BIM components and crane operations facilitates a seamless simulation of lifting paths, significantly reducing the likelihood of on-site errors. The integration of crane charts supported decision-making by providing clear visualizations of operational feasibility for crane setups of both 180 t and 300 t capacities. Multiple case studies assessed the impacts of crane capacity, site accessibility, and lifting height on crane performance. Simulations were conducted for both 180 t and 300 t cranes to compare costs and operational efficiency. In one case study, a four-span precast girder bridge with

prefabricated decks was selected as the model project. The project featured two parallel bridges: an inbound bridge and an outbound bridge, each comprising four spans with a maximum span length of 50 m. Figure 11 illustrates the project layout, detailing the positioning of the precast girders and prefabricated decks across the four spans. This diagram provides a comprehensive overview of crane operations, including the lifting stages and crane positions used during deck installation. This project layout served to apply and validate the Digital Twin methodology with an emphasis on optimizing crane operations for prefabricated deck erection.

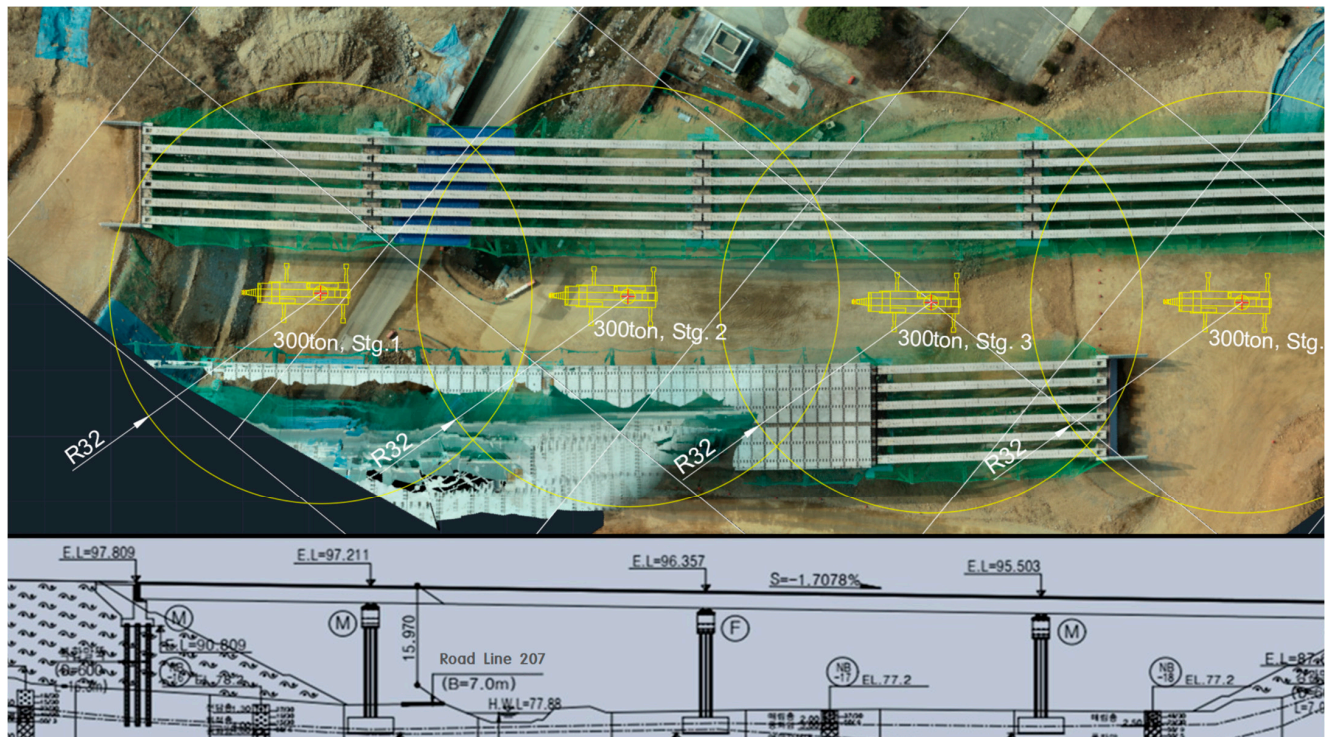


Figure 11. Project layout of a prefabricated bridge.

Table 5 details the calculated weight of the prefabricated deck. This weight is determined by multiplying the unit weight of concrete by the deck's volume and adding the weight of the lifting frame. The combined weight of the prefabricated deck and lifting frame constitutes the total weight of the lifting object. Additionally, to ensure safety, a safety margin is applied by multiplying the calculated total weight by a safety factor or by maintaining a safety margin relative to the crane's lifting capacity. For the verification of safety margins or efficiency, refer to Table 6.

Table 5. Calculation of the weight for the prefabricated deck.

Description	Contents	Remarks
Dimension and Weight (1)	$4.093 \times 2.5 = 10.233 \text{ t}$	Deck Width 9.0 m
Dimension and Weight (2)	$2.870 \times 2.5 = 7.175 \text{ t}$	Deck Width 6.5 m
Weight of Lifting Frame	2.527 t	H-Beam, Length = 10 m
Total Weight (1)	12.760 t	Deck (9.0 m) + Lifting Frame
Total Weight (2)	9.700 t	Deck (6.5 m) + Lifting Frame

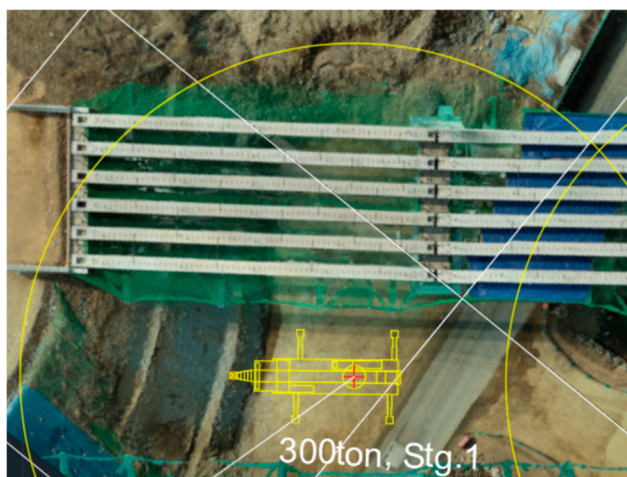
To determine the crane's capacity, three parameters—lifting weight, boom length, and working radius—were assessed. The lifting weight is calculated as shown in Table 5. The site conditions dictate the measurements for the boom length and working radius.

Practically, a circle can be drawn on a site map to visualize the maximum working radius, with the boom length determined by the elevation difference relative to the working radius. Table 6 illustrates the method for determining crane capacity based on weight, boom length, and working radius, providing insight into the safety margins of crane lifting capacity.

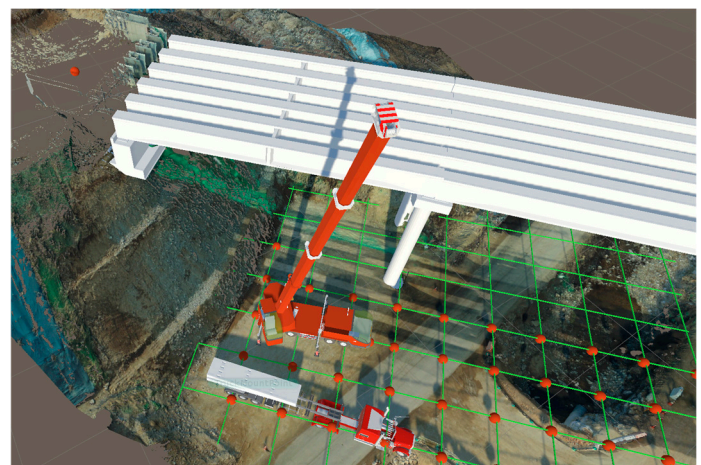
Table 6. Determination of crane capacity.

Capacity of Crane	Working Radius	Boom Length	Lifting Weight	Object Weight	Efficiency	Remarks
300 t	26 m	43.9 m	19.2 t	12.760 t	66.5%	Deck (1)
	32 m	58.5 m	13.3 t	9.702 t	72.9%	Deck (2)

Figure 12 depicts the planning of the prefabricated deck erection using a 300 t mobile crane. The yellow circle represents the crane’s working radius, with the crane’s lifting position marked at the center of the circle. The radius of this circle varies according to the lifting capacity of the crane. In comparison, a simulation was conducted using the CDT at the same lifting position with a mobile crane of identical capacity.



(a) Deck erection plan in method statement



(b) Deck erection simulation using Digital Twin

Figure 12. Comparison of deck erection plans.

4.2. Simulation Setup and Scenarios

A case study was conducted to evaluate the feasibility of using a smaller capacity crane compared to the original plan, which had specified a 300 t crane. Through simulations using the CDT, a 180 t crane was tested as a potential alternative. While a 180 t crane could not complete all deck erections from a single setup, the tasks could be accomplished by repositioning the crane. The simulation focused on cost evaluation and optimization of crane capacity and repositioning in the abutment 2 area of the upper bound. The objective was to address the “Cost Optimization Problem” by strategically organizing crane position and capacity optimization beforehand. Typically, shifting a crane takes about 1.5 h, a duration that must be factored into the total erection time due to multiple crane shifts. For completing one span, a 300 t crane takes 2.4 days, while a 180 t crane takes 2.6 days. However, as cranes are scheduled daily, the required deployment time for any crane is effectively rounded to 3 days. Despite the additional time required for repositioning, the simulation demonstrated a significant cost reduction of 27% when utilizing the smaller-capacity crane, thereby achieving a balance between time and economic efficiency. Figure 13 illustrates the crane-shifting positions of the smaller-capacity crane.

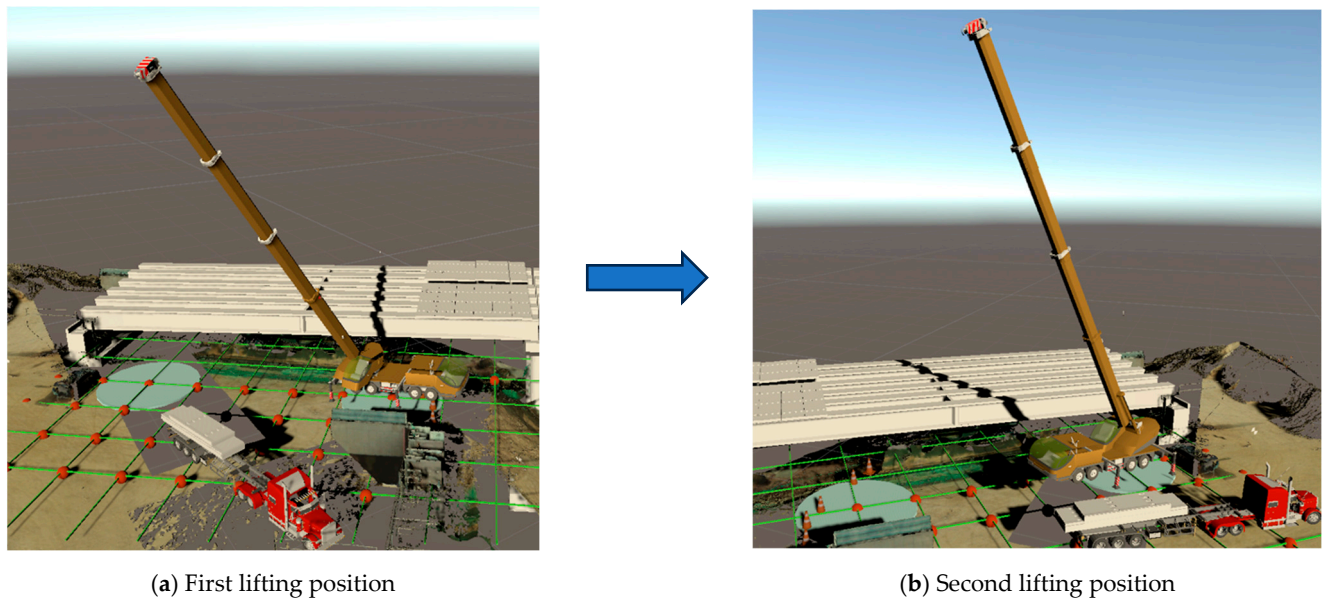


Figure 13. Crane shifting the lifting position with a smaller capacity crane.

To compare the costs incurred for prefabricated deck erection, the total cost can be calculated by multiplying the resources by their unit prices, as detailed in Table 7 for the 300 t crane and Table 8 for the 180 t crane. In this study, calculations were limited to crane rental charges and manpower costs. Given the equal duration of crane deployment (3 days) and identical manpower requirements across the scenarios, the results indicated substantial cost savings when using a smaller-capacity crane. For instance, the use of the 180 t crane resulted in a 27% cost reduction compared to the 300 t crane. Tables 7 and 8 provide a detailed cost breakdown for each crane deployment. The cost reductions achieved through optimization were calculated based on the actual costs incurred.

Table 7. Cost calculation for 300 t crane.

No	Item	Unit Price (USD/Day)	Quantity	Cost for Each Item (USD)
1	Mobile Crane (300 t) - Operator included	4286	3 day	$4286 \times 3 = 12,858$
2	Foreman: 1-person	146	3 man-day	$146 \times 3 = 438$
3	Semi-Skilled Labor: 2-person	116	6 man-day	$116 \times 6 = 696$
4	Unskilled Labor: 2-person	116	6 man-day	$116 \times 6 = 696$
5	Rigger: 2-person	201	6 man-day	$201 \times 6 = 1206$
Total Cost				15,894

4.3. Simulation for DFA: Optimizing Deck Dimensions

The design for assembly (DFA) simulation explored how varying dimensions of prefabricated decks impact crane operations and construction efficiency, providing essential feedback to optimize both the design and installation processes. Utilizing CDT technology enhances DFA by facilitating simulations that refine the dimensions of prefabricated decks. In this case study, three deck widths—2.0 m, 2.2 m, and 2.4 m—were evaluated using the CDT to identify the most efficient size based on crane capacity and lifting positions.

The trade-offs between prefabrication design decisions and crane operational efficiency have been carefully analyzed in this study, focusing on the critical parameters influenc-

ing both design and operational aspects. The prefabricated deck design must address constraints arising from the deck's dimensions, transportation, and crane handling.

Table 8. Cost calculation for 180 t crane.

No	Item	Unit Price (USD/Day)	Quantity	Cost for Each Item (USD)
1	Mobile Crane (180 t) - Operator included	2857	3 day	$2857 \times 3 = 8571$
2	Foreman: 1-person	146	3 man-day	$146 \times 3 = 438$
3	Semi-Skilled Labor: 2-person	116	6 man-day	$116 \times 6 = 696$
4	Unskilled Labor: 2-person	116	6 man-day	$116 \times 6 = 696$
5	Rigger: 2-person	201	6 man-day	$201 \times 6 = 1206$
Total Cost				11,607

The bridge width is determined during the basic design phase, leaving the deck width along the bridge's longitudinal direction as the primary decision variable during the detailed design stage. Deck weight, influenced by its width, directly affects operational efficiency and transportation logistics. Regulatory constraints on transportation play a pivotal role. According to vehicle operation restrictions, axle loads are limited to 10 t, and the total vehicle weight is capped at 40 t. After accounting for the vehicle's tare weight, the deck weight is effectively limited to below 25 t. If the deck weight exceeds 12.5 t, only one deck can be transported per trailer trip, reducing transport efficiency. Therefore, optimizing the deck weight within these limits is essential for balancing design and logistical efficiency. Regulatory restrictions also impose width and height limits for transport, which are set at 2.5 m and 4 m, respectively. These constraints necessitate careful design to ensure compliance while maximizing deck dimensions for efficient construction.

To explore the trade-offs, this study analyzed deck widths of 2.0 m, 2.2 m, and 2.4 m, considering their impact on crane operational efficiency and transportation logistics. The sensitivity analysis revealed that increasing deck width improves construction efficiency by reducing the number of lifts required. However, wider decks also result in heavier loads, potentially requiring higher-capacity cranes, which could reduce overall operational efficiency. The balance between deck dimensions and crane capacity thus becomes a critical trade-off in optimizing prefabrication design and operational logistics.

As deck width increases, so does its weight, which directly influences the working radius and lifting capacity of the crane. Integrating DFA principles, this study aimed to synchronize crane operations with prefabrication processes to enhance on-site assembly efficiency and overall productivity. The simulation results provided insights into deck dimension optimization, ensuring that both crane operations and prefabrication processes were streamlined. This alignment between the prefabricated components and crane capabilities led to significant enhancements in construction workflows.

4.3.1. DFA and Prefabricated Deck Design

The test site design specified a deck width of 2.0 m along the road direction. To examine the effects of different dimensions on crane operation and assembly, three deck configurations were prepared using the BIM algorithm: widths of 2.0 m (standard), 2.2 m, and 2.4 m. Figure 14 displays the prefabricated decks with varying widths. As the width of the deck increases, its weight also rises proportionally, affecting the working radius and overall lifting capacity of the crane. Therefore, selecting the optimal deck dimensions requires a careful balance among the weight, required boom length, working radius, and crane

capacity. After preparing the design options for the prefabricated decks, a CDT simulation was conducted to analyze the erection performances of these varying dimensions.

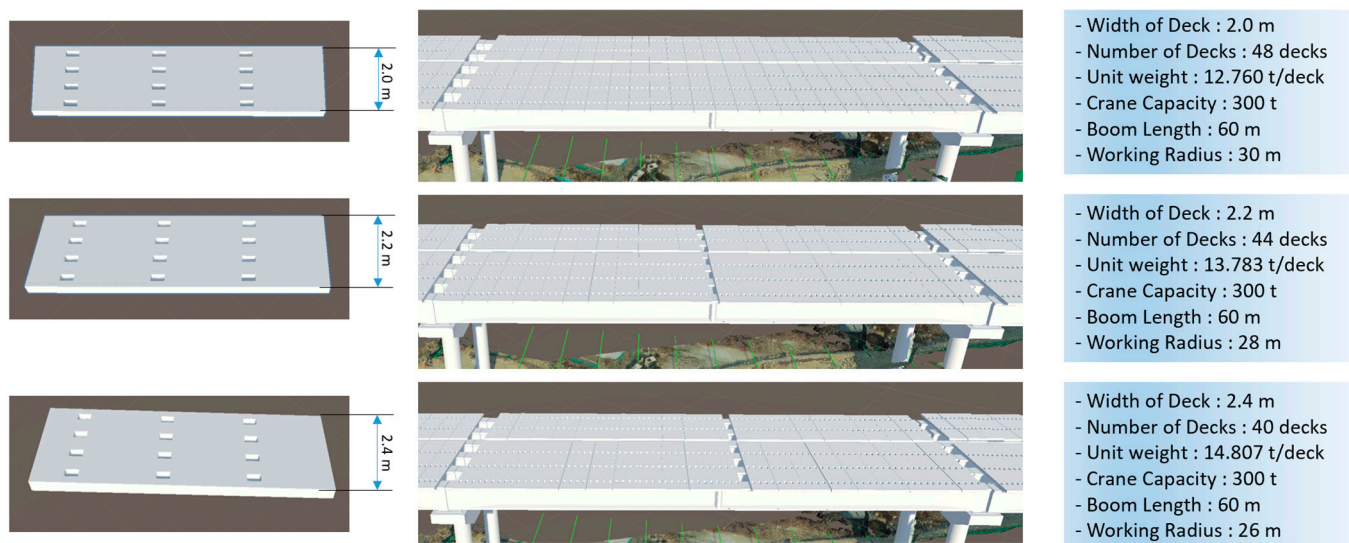


Figure 14. Prefabricated decks with varying widths.

4.3.2. CDT Simulation with Different Dimensions Decks

The CDT technology was utilized to simulate a crane's capability to lift prefabricated decks from various grid positions on a construction site. The simulation evaluated the number of decks that could be lifted by cranes of specific capacities, considering grid positions and deck dimensions. Each simulation run accounted for the crane's maximum reach, working radius, and the number of decks it could safely lift from different positions.

Table 9 summarizes the simulation input parameters for different deck widths and compares the number of decks that can be lifted by cranes with specified capacities, alongside corresponding deck weights and crane configurations.

Table 9. Simulation input parameters for 300 t crane.

Description	W 2.0 m	W 2.2 m	W 2.4 m	Remarks
Deck Weight	18.229 t	19.690 t	21.153 t	70% Safety Margin
Boom Length	59.4 m	59.4 m	59.4 m	
Working Radius	30 m	28 m	26 m	

In this research, the deck dimensions along the bridge's horizontal alignment are identical; therefore, the weight for each deck component can be calculated proportional to the width. Figure 14 shows the prefabricated decks with varying widths.

The results of the simulation indicated that decks with a W 2.0 m width required fewer crane adjustments, facilitating faster erection times. Conversely, while the W 2.4 m deck width proved more efficient in reducing the total number of components, it presented challenges due to the increased weight, impacting the crane's lifting capacity. These findings underscore the trade-offs between assembly efficiency and crane operational constraints when varying deck dimensions. The strategic integration of DFMA principles within the prefabricated bridge deck design process ensures that crane operations and prefabrication processes are cohesively aligned for optimal performance. By leveraging CDT simulations, this study demonstrates that adopting a slightly wider deck of W 2.4 m under specific grid conditions can enhance assembly efficiency compared to the standard W 2.0 m. However,

this adjustment necessitates careful consideration of the crane's capacity to ensure safe and effective lifting operations.

4.3.3. Comparison and Strategic Decision-Making

Although the lifting and target erection positions remained constant, the dimensions of the deck resulted in varying lifting weights. Consequently, the working radius of the crane adjusted according to the lifting weight, affecting performance levels. The simulation results elucidate how deck weight and grid position influence the number of decks lifted, which can be interpreted as a measure of the performance of crane erection activities. In comparing each simulation case, grid point G(1,4) demonstrated maximum performance regardless of the deck dimensions. Notably, the W 2.4 m configuration proved to be the most efficient in terms of the total time required for the completion of one span erection. However, for both W 2.2 m and W 2.4 m, not all decks within a span could be completed in a single shift due to the limited working radius of the crane, leaving a few decks unfinished. This limitation necessitated additional efforts to complete the erection. However, by fine-tuning the crane-lifting position, erection performance was significantly improved. Fine-tuning was specifically implemented to enhance the performance of W 2.4 m deck dimensions. In this instance, the grid interval was adjusted from 5 m to 1 m. Figure 15 displays the adjusted grid settings.

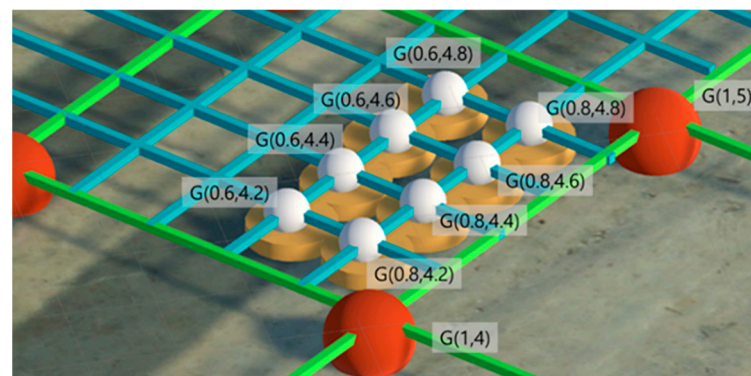


Figure 15. The adjusted grid setting with narrower intervals.

Deck-erection simulations were conducted at precise grid points. In the initial simulation under the 5 m interval grid, deploying the wider deck design appeared to offer little benefit. In these cases, the W 2.0 m width emerged as the most balanced option, requiring fewer crane adjustments and allowing for a greater number of decks to be lifted compared to the other widths. However, by adjusting the grid interval only between G(1,4) and G(1,5) to a 1 m interval and retrying the simulation, the results revealed that a wider deck dimension of W 2.4 m was the most efficient during the adjusted grid interval simulation. Therefore, by incorporating this feedback into the design of the deck, the total construction cost can be reduced, potentially leading to more effective construction practices. Figure 15 shows the adjusted grid setting with narrower intervals for fine-tuning.

This strategic decision-making process, supported by CDT simulations, aims to optimize both crane operations and prefabrication processes. The results of this study align with the findings of Nguyen et al. (2024) [29], which underscore the importance of integrating DFMA principles with BIM-based simulations to enhance decision-making in prefabricated construction. Their study highlighted how preassembly analysis could proactively address assembly errors and optimize crane operations during the planning phase. Table 10 shows the simulation results.

Table 10. Simulation result for adjusted grid settings.

Description	Number of Decks Erected	Remarks
G(1,4)	39 decks	Applied on the 5 m × 5 m grid position
G(0.8,4.2)	40 decks	One span erection completed
G(0.8,4.4)	40 decks	One span erection completed
G(0.8,4.6)	39 decks	Deck Z2 excluded
G(0.8,4.8)	38 decks	Deck Z1 and DeckZ2 excluded
G(0.6,4.2)	19 decks	collusion with outer girder
G(0.6,4.4)	40 decks	One span erection completed
G(0.6,4.6)	40 decks	One span erection completed
G(0.6,4.8)	38 decks	Deck Z2 and DeckZ40 excluded
G(1,5)	37 decks	Applied on the 5 m × 5 m grid position

4.3.4. Cost Analysis of DFA Case Study

To determine the optimal deck dimensions, it is crucial to not only assess the number of decks that can be installed but also to conduct a comprehensive cost analysis. In this research, the total cost of the DFMA case study was calculated, encompassing three major components: manufacturing, transportation, and installation. The cost analysis revealed that as the width of the deck increases, the total cost tends to decrease. However, this trend holds true only if a crane of the same capacity is utilized. Should a higher-capacity crane become necessary due to increased deck weight, the cost analysis would require recalibration. Fabrication costs include expenses related to formwork, materials such as rebar and concrete, labor, and other miscellaneous items. The cost of formwork is directly influenced by the deck dimensions, as the required formwork area increases proportionally with the deck width. Regarding fabrication costs, while formwork expenses do increase with deck dimensions, their impact is mitigated by a high rate of reuse, assumed here to be 100 times.

The costs of deck fabrication, transportation, and installation may vary due to price fluctuations; however, a sample cost analysis can be conducted based on assumed prices for each cost variable. Table 11 presents a comparison of the construction costs, encompassing fabrication, transportation, and installation. As the deck width expands, fabrication costs rise; however, the financial impact remains minimal due to the high reuse rate of form sets. Conversely, transportation costs are affected by the number of decks, while installation costs depend on the crane's capacity and operational duration. Since crane rental costs are typically charged on a daily basis, particularly for heavy-duty cranes, employing a crane of the same capacity for wider deck dimensions results in a lower total cost.

Table 11. Cost summary for each deck dimension.

Description	Deck No.	Formwork Area (m ²)	Fabrication (USD)	Transportation (USD)	Installation (USD)	Total Cost (USD)
W 2.0 m	25	23.5	75,118	9000	22,742	106,860
W 2.2 m	22	25.4	71,627	8000	22,742	102,369
W 2.4 m	20	27.3	70,137	7000	18,456	95,592

4.4. Results

This section presents case studies utilizing the CDT to optimize crane operations during the erection of prefabricated bridge decks. By simulating scenarios with varying lifting heights, accessibility conditions, crane capacities, and deck dimensions, the CDT identified optimal lifting positions and strategic alternatives based on cost and performance.

In scenarios involving different lifting heights, the CDT adjusted the crane's boom length and working radius to enhance deck erection performance and reduce operational

costs. For accessibility constraints, the Digital Twin generated detailed 3D simulations that facilitated the selection of crane positions, balancing operational feasibility with site-specific limitations.

Simulations comparing the use of a 180 t crane with a 300 t crane demonstrated significant cost savings. Although the 180 t crane required additional repositioning, it achieved a 27% cost reduction compared to the 300 t crane, effectively balancing time efficiency and economic considerations. Simulations involving various deck dimensions showed that the W 2.4 m deck width offered the highest efficiency when combined with precise crane positioning.

These findings highlight the practical utility of CDTs in planning and optimizing construction operations, particularly for complex projects where site-specific conditions and equipment configurations significantly impact outcomes. The CDT provided a simulation environment that enhanced project execution efficiency, improved resource allocation, and reduced costs. Additionally, it served as a valuable tool for design refinement, evaluating different deck width options, improving assembly performance, and mitigating operational risks.

5. Discussion

5.1. Summary of Key Findings

This research successfully developed a Construction Digital Twin (CDT) framework to optimize crane lifting positions and capacities for the erection of prefabricated bridge decks. The methodology demonstrated significant improvements in cost reduction, operational efficiency, and safety by integrating 3D site modeling, BIM manipulation, and systematic crane operation simulations. Using an exhaustive search method, the optimal crane lifting positions and configurations were identified, reducing the need for crane repositioning and minimizing rental costs. The results further showed that even smaller-capacity cranes, when optimally configured, can achieve comparable performance to larger cranes, offering cost-effective alternatives for construction projects.

5.2. Comparison with Existing Literature

These findings align with previous studies that underscore the potential of Digital Twins in improving construction efficiency, such as those by Nguyen et al. (2024) [29] and Ghasemi et al. (2024) [30]. Unlike studies that rely extensively on advanced AI or ML technologies—which are computationally intensive and expensive—this research employs a practical and accessible approach using exhaustive search techniques. This method makes the CDT particularly well-suited for small- and medium-scale projects, addressing a gap often overlooked in earlier studies that prioritized large-scale, high-tech implementations.

Ghasemi et al. (2024) [30] proposed an approach for optimizing multi-tower crane locations through an Application Programming Interface (API) integrated with Building Information Modeling (BIM). Their research demonstrated notable reductions in time and cost during prefabricated construction. However, the focus of their study was limited to tower crane operations, which are typically static and confined to predefined positions. In contrast, this study addresses the dynamic nature of mobile crane operations, incorporating considerations such as site constraints, crane lifting positions, and operational safety. These elements, which are critical for optimizing the erection of prefabricated bridge decks, are not fully explored in their research. Similarly, Gwak et al. [21] developed a genetic algorithm (GA)-based optimization method for mobile crane repositioning route planning. Their study primarily focused on minimizing repositioning time and optimizing crane movement paths. While effective for route planning, their approach did not address lifting positions, the integration of BIM with crane operations, or site-specific constraints, such as

obstacles and working radii. This research overcomes such limitations by integrating 3D site models and BIM with crane simulations to evaluate and optimize crane lifting positions dynamically. The incorporation of collision detection further enhances the realism and safety of the proposed methodology.

Furthermore, integrating DFMA principles within the CDT has been proven to enhance the planning process for prefabricated components. These findings align with the work of Salimi et al. (2018) [31], which demonstrated the positive impact of DFMA on manufacturing efficiency. This research extends these principles specifically to crane operations, creating a link between component design and on-site assembly and thereby optimizing crane lifting strategies.

5.3. Practical Implications

The practical implications of this research are significant, particularly for small- and medium-sized construction firms seeking cost-effective methods for crane operation optimization. By leveraging a CDT framework that integrates 3D site modeling, BIM, and crane simulations, this study provides an accessible solution that does not rely on advanced AI or IoT technologies. The interactive platform enables construction planners to make informed decisions during the planning phase, reducing unnecessary crane shifts, minimizing rental costs, and enhancing overall project efficiency. The scalability and adaptability of the CDT make it a useful tool for various construction scenarios, including those with constrained resources or complex site conditions.

5.4. Limitations and Future Research

While the CDT framework demonstrates robust capabilities, there are several limitations that warrant further investigation. First, the exhaustive search method, although effective for small and medium-scale sites, becomes computationally demanding for larger and more complex construction environments. Future studies should explore heuristic or metaheuristic algorithms to enhance computational efficiency without compromising accuracy. Second, the simulations conducted in this study are based on static site conditions. The integration of real-time data through IoT sensors could significantly enhance the CDT's applicability by enabling dynamic adaptations to evolving construction site conditions. Incorporating real-time feedback into crane operations could also improve planning accuracy and operational responsiveness. Finally, while this research focused primarily on prefabricated bridge decks, future work could extend the CDT framework to other prefabricated components, such as girders and columns. This expansion would provide a more comprehensive evaluation of CDT's utility across diverse construction applications, contributing to broader adoption and implementation in the industry.

6. Conclusions

6.1. Summary of Findings

The role of cranes in construction has become increasingly significant with the advancement of off-site construction methods. As crane deployments have become more frequent, optimizing crane operations has emerged as a critical factor for cost control in the handling and installation of prefabricated components. Current crane operations often lack an optimal approach due to prioritizing safety concerns. In situations where crane deployment is limited and critical, safety tends to be emphasized over operational optimization. Conversely, when crane operations become central to construction activities, overestimating crane capacity may hamper cost control. These challenges highlight the need for an optimized approach to crane deployment that balances safety and efficiency. Addressing this need forms the core problem explored in this research.

This research successfully developed a Construction Digital Twin (CDT) system to optimize crane operations for prefabricated bridge deck erection. By integrating UAV-based 3D site modeling, Building Information Modeling (BIM), and crane simulation using the Unity engine, the CDT enhances crane positioning and capacity management. The framework provides a scalable and practical solution that addresses real-world challenges in construction. The integration of DFMA principles within the CDT further enhanced the planning process by linking component design in terms of dimensions with crane operation strategies, improving assembly efficiency. These findings contribute to the growing body of evidence supporting the potential of DT technologies to improve construction processes, with a particular focus on making these technologies accessible and practical for various prefabricated members' erection projects.

6.2. Contributions to the Field

This study makes several key contributions to the field of construction planning and Digital Twin (DT) technology. A Construction Digital Twin (CDT) framework was developed to optimize crane position and capacity, addressing three main types of constraints: accessibility constraints, installation position constraints, and equipment constraints. For accessibility constraints, the research proposed and implemented a methodology to create a 3D site model using UAV data, effectively addressing challenges unique to specific construction sites. To ensure prefabricated deck installation at designated positions according to the required sequence, referred to as installation position constraints, the bridge BIM model was integrated with the 3D site model. This integration incorporated collision-detection capabilities, enabling operational safety and precision when working with erection equipment such as cranes and trailers. Equipment constraints were addressed by developing a crane operational methodology that included motion simulation and modeling for lifting and installing prefabricated decks. These individual components were synthesized into a cohesive CDT framework that simulates construction processes and offers an enhanced approach to construction planning. This framework demonstrates its utility as a design for assembly (DFA) tool, surpassing conventional methodologies.

The study also introduced a novel approach to dynamic BIM manipulation. Unlike traditional BIM applications, which focus primarily on static design and visualization, this research enabled the dynamic separation and manipulation of prefabricated bridge decks and other structural elements. These components were coordinated with crane operations during construction simulations, closely mirroring real-world processes. Additionally, an algorithm was developed to align the geometry and pivot centers of prefabricated decks, ensuring seamless transportation, lifting, and installation. This capability marks a significant advancement in the integration of BIM manipulation into construction workflows. A further contribution of this research involved enhancing the accuracy of crane operation simulations by incorporating real-world data. Parameters such as lifting and idle times were extracted through frame-by-frame analysis of video footage from an actual deck erection site. These parameters were applied to the crane model within the CDT, allowing for realistic simulations of crane operations. The timescale of these simulations was accelerated by 100 times, enabling faster acquisition of results without compromising precision.

The optimization of crane-lifting positions was another significant contribution. The study implemented an exhaustive search methodology to identify optimal crane positions. Initially, 5 m grid was applied to the 3D site model, which was subsequently refined to 1 m intervals for more detailed analysis. This iterative process enhanced the understanding of site conditions and allowed for precise identification of crane positions, facilitating informed decision-making. The integration of 3D site modeling, BIM, and crane operations created a realistic simulation environment, improving safety and operational efficiency in

prefabricated deck erection planning. The CDT framework also demonstrated its potential as a tool for design for assembly by simulating the effects of varying deck dimensions. Through these simulations, the research showed how different prefabricated deck widths could impact assembly performance and operational risks. By incorporating detailed terrain modeling, dynamic BIM manipulation, and crane operation simulations, the study delivered practical advancements in construction planning and execution.

In conclusion, this research makes notable contributions to Digital Twin technology in construction. By presenting a practical framework for optimizing crane operations, it enhances the precision and safety of construction planning processes. The integration of dynamic BIM manipulation with 3D site modeling represents a significant step forward in advancing construction methodologies. Furthermore, the CDT has demonstrated its utility as a DFA tool by enabling evaluations of different prefabricated deck dimensions. These findings provide valuable insights into improving assembly performance, mitigating operational risks, and fostering innovation in construction planning.

6.3. Future Research Directions

The subsequent research will be implemented based on this research and extended focusing on the girder erection activities. Site logistics for the girder will be investigated to propose the most efficient site routes. This approach would assist in planning crane paths and trailer movements, minimizing delays, and improving overall efficiency. Furthermore, deploying two cranes simultaneously for precast girder erection could be studied to evaluate the effectiveness of dual-crane operations in enhancing lifting stability and reducing installation time. In addition, future research should explore the integration of real-time IoT sensors for monitoring crane operations, which could provide dynamic updates to the CDT and improve the accuracy of simulations during ongoing construction projects. Real-time data from IoT sensors could enable adaptive simulations that respond to changing conditions on-site, further enhancing decision-making for operational safety as well as cost reduction.

The CDT can also be a framework for machine guidance/machine control toward the fully automated unmanned equipment operation. Based on the CDT framework, not only IoT sense or monitoring tools but also automatic equipment operation controllers can be equipped for fully automated unmanned equipment operation. Future research involves studying the potential benefits of incorporating machine learning for predictive analysis to further optimize crane operations and enhance decision-making processes. This could include predictive maintenance of crane equipment and predictive scheduling to anticipate potential delays or risks. The emphasis would be on providing practical, data-driven solutions without over-relying on advanced AI technologies, thus maintaining accessibility for a wider range of construction projects.

The proposed future research directions aim to build upon the foundational CDT framework developed in this study, expanding its applicability to different types of construction activities and incorporating advanced data-driven tools where beneficial. The focus remains on practical implementation, ensuring that the technologies and methodologies are scalable and accessible for real-world construction projects.

Author Contributions: Conceptualization, D.-H.J.; methodology, D.-H.J., G.-T.R., C.-H.J.; validation, D.-H.J.; investigation, D.-H.J. and G.-T.R.; resources, C.-S.S., C.-H.J. and G.-T.R.; writing—original draft preparation, D.-H.J.; writing—review and editing, C.-S.S. and C.-H.J.; supervision, C.-S.S.; project administration, C.-S.S.; and funding acquisition, C.-S.S. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Korea Agency for Infrastructure Technology Advancement under the Ministry of Land, Infrastructure and Transport, and managed by the Korea Expressway Corporation grant number [RS-2020-KA156007].

Data Availability Statement: All data, models, and code generated or used during the study appear in the published article.

Acknowledgments: Research for this paper was conducted with the support of the National R&D Project for Smart Construction Technology.

Conflicts of Interest: The authors declare no conflicts of interest.

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